

CONTROL INFORMATION UNDER STATE CONVERGENCE: ENTROPY, PRECISION AND RELIABILITY

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ABSTRACT. We study the information required to maintain a state constraint when every feasible state trajectory converges to a common point. For smooth systems on solid cones, we prove exact finite-horizon transference to a normalized affine system and spectral formulas for exact, outer, and reference-trajectory tracking invariance entropies. Their planar formulas persist under smooth perturbations with fixed first jet. Subexponential radial decay gives equal positive control entropies, although the state-trajectory entropy, input-fibre entropy, and conditional entropy of every invariant lift measure vanish. We then consider binary planar systems contracting toward the origin under every input and let the Euclidean tolerance decrease as $\exp(-\gamma n + o(n))$. Matched first derivatives determine the complete full-state precision spectrum and a strict cost difference from one fixed nearly full-area compact initial set; unequal radial powers produce this difference already at small positive precision. At failure probability $\exp(-\alpha n + o(n))$, nonsingular derivatives on the constraint give the rate $\min\{S(\gamma), L_\alpha(\alpha)\}$, where L_α is the classical binary source-coding reliability function. An injective example with null critical sets preserves every full-state catalogue count but changes the reliability cost. The proofs connect arbitrary approximate inputs to actual covered state regions and physical area, rather than presupposing an optimal symbolic coding.

1. INTRODUCTION

Invariance entropy counts the control signals that must be available to maintain a prescribed state constraint. It measures a choice problem over initial states, rather than the orbit complexity of one autonomous state map. The distinction matters when viable trajectories all converge: increasingly precise control choices can still be required to prevent departure. We examine this information cost at three levels—exact viability, a fixed environmental tolerance, and a tolerance and failure probability that decrease with the horizon.

The common object is a finite catalogue of open-loop inputs. The catalogue is fixed before the initial state is selected; its assigned member may depend on that actual state. An approximate input must satisfy the Euclidean distance constraint at every intermediate time. Leaving and reentering the constraint does not remove an earlier violation. For probability costs, success is measured using uniform physical area on a fixed triangle; a horizon-dependent success set is different from one fixed compact initial set of nearly full area.

1.1. Two levels of convergence and their information costs. Our first results concern smooth invertible controlled maps on a full-dimensional solid cone. A common radial motion forces every feasible trajectory to approach the apex, while normalized transverse motion can expand. Theorem 3.1 transfers the exact finite-horizon catalogue to an affine system and computes outer and tracking rates on every positive-volume compact initial set. Theorem 3.2 preserves the planar rates under an infinite-dimensional class of smooth perturbations with fixed first jet. With radial map arsinh , all three control entropies can

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be positive and equal. The common Lyapunov theorem, Theorem 7.1, identifies the separation from state complexity: the state-trajectory shift and input fibres have zero entropy, and every invariant lift measure is supported on the fixed state. The input shift accounts for the entropy of the complete constrained lift. We also give arbitrary-dimensional finite-alphabet counts, prescribed exact and outer rates, and a feedback regularity obstruction.

The second group of results imposes the stronger requirement that every input contract toward the origin in a common norm, including trajectories outside the constraint. Fixed positive tolerance then has zero outer and tracking entropy, as Proposition 8.1 records. Shrinking the tolerance reveals information not seen by that limit order. For binary planar maps with matched first derivatives, Theorem 9.3 gives the complete spectrum

$$S(\gamma) = \max_{p \in [0,1]} H_b(p) \min\{1, \gamma/[\beta\chi(p)]\}.$$

All positive-area compact sets share the rate γ/β up to βh . A single compact set of arbitrarily nearly full area has rate $\min\{h, \gamma/\beta\}$ at every finite precision exponent, whereas the full-triangle spectrum grows to $\log 2$. Within the stated first-order matrix family, unequal radial powers separate the two costs already near zero (Theorems 9.6 and 9.5). These conclusions allow nonlinear feedback and folds; null critical sets suffice for their finite-transient area argument.

Exponential reliability requires a stronger comparison. Theorem 9.1 assumes no critical points on the triangle and obtains the rate $\min\{S(\gamma), L_a(\alpha)\}$ at failure exponent $\alpha > 0$. The function L_a is classical; the assertion is its validity for physical area and arbitrary approximate controls after a nonlinear transient. The proof transports probability through all feasible transient branches and constructs disjoint positive-area regions that force every successful input to retain specified prefixes. Theorem 9.2 gives an injective C^2 example with null critical sets, no positive-area collapse, and unchanged full-state counts at every horizon and tolerance, but a strictly larger reliability cost. Thus preservation of ordinary precision costs does not imply preservation of a reliability exponent.

The two groups are connected by their catalogue problem, not by an inclusion of all hypotheses. In the cone family, convergence is imposed on feasible states and unsuccessful transverse errors may expand. In the planar precision family, contraction is global under every input. The former also allows subexponential radial decay and arbitrary dimension. The latter has specified binary first-order geometry and a triangular constraint. No theorem identifies these classes by a singular normalization at the apex or extends a conclusion beyond its own assumptions.

1.2. Relation to earlier work. Invariance entropy was introduced and developed in [2, 3, 4, 32]. Exact, strict, outer, and strong terminology varies across sources; our tracking definition is the discrete-time counterpart of the reference-trajectory quantity in [3, Definition 3.1.1]. Linear Bowen coverings, spectral expressions, and unstable-volume estimates are established tools [3, 5]. We do not claim a new volume method or novelty for an expanding determinant. Stabilization and network data-rate problems [29, 10, 11] optimize related costs for prescribed objectives. Our first family instead makes every feasible trajectory converge, independently of a selected stabilizing controller.

Control-set and hyperbolic formulas [6], partially hyperbolic bounds [7], and the discrete hyperbolic theory [8] are conceptual precedents for distinguishing control information from state complexity. Their full-time lifts and controllability or hyperbolicity assumptions do not directly give the forward-cone counts here. The cones are not approximately controllable control sets, and their full-time state core is the apex. The neutral radial derivative in the equal-entropy examples also lies outside a hyperbolic splitting without centre. Yao's preprint [9] already provides positive exact entropy on a thin constraint with zero approximate entropy; its feasible state coordinates have limiting pair distances. Our contribution is not the weak assertion that positive exact entropy can coexist with an absence of state

Li–Yorke pairs. The solid-cone calculation, simultaneous positive outer and tracking rates, and specified nonlinear persistence are the additional statements.

Positive state entropy has established consequences for chaos [1]; relative and random entropy theories likewise use their stated state or conditional entropy hypotheses [20, 21]. We calculate the actual state factor and input fibres instead of inferring those hypotheses from a control catalogue. Affine homogenization is familiar [12], but the coordinate $y = z/s$ deletes the apex and is not a compact-constraint conjugacy. The Lie-group representation obstruction to applying [13] directly is stated in Remark 5.2. Equi-invariability and invariance complexity [14, 15, 16] describe nearby states and finite input families; departure of a nearby state is compatible with convergence of all states that remain feasible. The measurable data-rate and entropy-dimension theories [17, 18, 19] concern other optimizations and growth scales, rather than the conditional lift entropy computed here. Recurrence catalogues [24] and nonautonomous skew products [25] supply further context without identifying control cost with state-trajectory entropy.

For the planar precision results, the finite-horizon full-cover object is outer invariance complexity [16, Section 2.2]; that framework already permits noninjective continuous maps. Wang, Huang and Sun [26] use invariant-partition cylinder probabilities, and their conjugacy result transports the initial measure with the system. Colonius [28] relates quasi-stationary metric invariance entropy to coder-controllers. Wang and Huang [33] study spanning-based measure entropy and inverse variational principles. Probability-dependent control entropy and measure transport are therefore not new in themselves. Nie, Wang and Huang [27] obtain measure entropy by convex duality from invariance pressure; Chen, Huang and Zhong [19] use specified partition cylinders and time-power weights. Those descriptions do not by themselves settle coverage: application to our catalogue still requires quantitative control of the sets actually covered by arbitrary approximate inputs. The complete-region estimates, all-branch probability transport, and physical forcing below provide that relation. Uniform physical area remains fixed; a transported-measure identity alone does not give this comparison.

Graph estimates, summable distortion, product stopping scales, typical sets, and source-coding reliability are classical tools [7, 31, 30]. Cost-threshold pressure and partition formulas are developed in [22, 23]. Our type optimisation is an elementary application of these tools. The extra control arguments concern complete physical feasible regions, arbitrary approximate inputs, and their mass under the original initial-state distribution. Appendix A makes explicit what symbolic counting proves and the realizability assumptions it requires.

1.3. Organization. Section 2 defines the entropy quantities and state factors. Sections 3–7 give the cone formulas, their complete proofs, finite-alphabet examples, and entropy separation. Section 8 explains the change from fixed tolerance to shrinking precision. Section 9 states the planar hypotheses and results; Sections 10–13 establish their physical geometry and precision costs. Sections 14 and 15 prove the reliability theorem and construct its obstruction. Sections 16 and 17 give the mismatch criterion and examples. The final section collects the consequences and scope. The appendices retain the precise partition-counting interface, a continuous-time scalar baseline, and comparisons for different control ranges.

2. CONTROL CATALOGUES AND STATE FACTORS

2.1. Control catalogues and state chaos. Let X be a metric space and let U be a nonempty compact metric space. Throughout the main text $f : X \times U \rightarrow X$ is continuous, $\mathcal{U} = U^{\mathbb{N}_0}$ has its product topology, and every sequence is admissible. The uniquely defined forward solution is

$$\varphi(0, x, u) = x, \quad \varphi(n+1, x, u) = f(\varphi(n, x, u), u_n).$$

No negative-time admissibility is presumed, even when individual maps $f_v = f(\cdot, v)$ are invertible. A nonempty compact set $Q \subset X$ is *controlled invariant* if

$$\forall x \in Q \quad \exists u \in \mathcal{U} \quad \forall n \in \mathbb{N}_0 : \varphi(n, x, u) \in Q.$$

With unrestricted concatenation this is equivalent to the one-step condition $\forall x \in Q \exists v \in U : f(x, v) \in Q$. We never use the term strong invariance to mean an unspecified strengthening of this statement. The different universal condition $\forall x \in Q \forall v \in U : f(x, v) \in Q$ is not assumed.

For a nonempty compact $K \subset Q$, an (n, K, Q) -spanning family is a finite set $\mathcal{S} \subset \mathcal{U}$ satisfying

$$(2.1) \quad \forall x \in K \quad \exists u \in \mathcal{S} \quad \forall k \in \{0, \dots, n\} : \varphi(k, x, u) \in Q.$$

The family is fixed before the initial state is selected; only its assigned member may depend on that state. Its minimum cardinality is $r_{\text{inv}}(n, K, Q)$, with value $+\infty$ if no finite family exists. Only the first n symbols matter. All logarithms are natural, and $\log(+\infty) = +\infty$. Define

$$(2.2) \quad h_{\text{inv}}(K, Q) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log r_{\text{inv}}(n, K, Q).$$

We call this *exact* invariance entropy; it is also called strict invariance entropy in some continuous-time sources. It does not require interior containment.

For $Q^\varepsilon = \{x \in X : \text{dist}(x, Q) < \varepsilon\}$, define

$$(2.3) \quad h_{\text{out}}(K, Q) = \lim_{\varepsilon \downarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \log r_{\text{inv}}(n, K, Q^\varepsilon).$$

The outer limit exists in $[0, +\infty]$ by monotonicity. The time limit is taken first. Neighborhoods are measured in the ambient state metric, not in a rescaled projective metric. In particular $h_{\text{out}}(K, Q) \leq h_{\text{inv}}(K, Q)$.

We also use the discrete-time counterpart of strong invariance entropy in [3, Definition 3.1.1]. A *tracking catalogue* consists of finitely many pairs (a_i, u^i) , $a_i \in Q$, whose reference trajectories are feasible in Q for all nonnegative times, and for which

$$\forall x \in K \quad \exists i \quad \max_{0 \leq k \leq n} d(\varphi(k, x, u^i), \varphi(k, a_i, u^i)) < \varepsilon.$$

Let $r_{\text{inv}}^+(n, \varepsilon, K, Q)$ be the least number of such pairs and put

$$h_{\text{inv}}^+(K, Q) = \lim_{\varepsilon \downarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \log r_{\text{inv}}^+(n, \varepsilon, K, Q).$$

Requiring reference viability only up to n gives the same number: a feasible finite reference trajectory can be continued from its endpoint by controlled invariance. The *actual* trajectory in this definition is required to track its reference, and need not itself stay in Q . Forgetting reference states gives

$$(2.4) \quad r_{\text{inv}}(n, K, Q^\varepsilon) \leq r_{\text{inv}}^+(n, \varepsilon, K, Q).$$

There is no general identification of tracking entropy with exact entropy in our definitions. Measure-theoretic invariance entropy, including partition and quasi-stationary versions [26, 27, 28], is another object. We do not calculate it by computing a conditional Kolmogorov–Sinai entropy.

A pair of feasible state trajectories x_n, y_n is a *state Li–Yorke pair* if

$$\liminf_{n \rightarrow \infty} d(x_n, y_n) = 0, \quad \limsup_{n \rightarrow \infty} d(x_n, y_n) > 0.$$

The same input may be imposed on both trajectories, or distinct inputs may be allowed. For a scrambled set with state-dependent open-loop inputs, a single assignment $x \mapsto u_x$ must first be fixed on the whole set; it cannot be reselected for each pair. A unified causal controller, including one with memory, gives another fixed rule. Our negative conclusion

quantifies over *all* pairs of feasible trajectories with arbitrary separate inputs, and hence covers all these more restrictive choices.

2.2. The true lift and its entropies. Let $\sigma : \mathcal{U} \rightarrow \mathcal{U}$ be the one-sided shift and define

$$(2.5) \quad \Lambda_Q = \{(u, x) \in \mathcal{U} \times Q : \varphi(n, x, u) \in Q \text{ for all } n \geq 0\}, \quad F(u, x) = (\sigma u, f(x, u_0)).$$

Each finite-time solution is continuous in (u, x) . Therefore Λ_Q is closed in the compact space $\mathcal{U} \times Q$, and F is a continuous self-map of Λ_Q . It is positively invariant; surjectivity does not follow. The input projection $\pi(u, x) = u$ satisfies $\pi F = \sigma \pi$. Its image need not in general be all of $\mathcal{U}$.

Use a bounded metric $d_U \leq 1$ on U , and the compatible input metric

$$d_{\mathcal{U}}(u, v) = \sum_{j=0}^{\infty} 2^{-j-1} d_U(u_j, v_j).$$

We equip the lift with the maximum of the input metric and the bounded state metric $\min\{1, d\}$. For a compact system $T : Z \rightarrow Z$, let $s_n(T, \varepsilon)$ denote the maximum cardinality of a set separated by more than ε in the Bowen metric $\max_{0 \leq k < n} d_Z(T^k z, T^k w)$. Then

$$h_{\text{top}}(T) = \lim_{\varepsilon \downarrow 0} \limsup_{n \rightarrow \infty} n^{-1} \log s_n(T, \varepsilon).$$

For nonempty input fibers use the same Bowen metric restricted to $\pi^{-1}(u)$, and define the uniform fiber growth rate

$$(2.6) \quad h_{\text{fib}}(F \mid \pi) = \lim_{\varepsilon \downarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \log \sup_{u \in \pi(\Lambda_Q)} s_n(F, \varepsilon, \pi^{-1}(u)).$$

This is the precise input-relative topological quantity used below. We do not use the informal subtraction $h_{\text{top}}(F) - h_{\text{top}}(\sigma)$, particularly when the latter is infinite.

The state-trajectory space is

$$\mathcal{T}_Q = \{(\varphi(k, x, u))_{k \geq 0} : (u, x) \in \Lambda_Q\} \subset Q^{\mathbb{N}_0}.$$

With its product topology and left shift S , it is a compact dynamical system: the trajectory map from Λ_Q is continuous and intertwines F and S . A compatible metric on $\mathcal{T}_Q$ is $\sum_{j \geq 0} 2^{-j-1} \min\{1, d(x_j, y_j)\}$. For an F -invariant probability measure μ , the conditional metric entropy used here is the usual $h_{\mu}(F \mid \pi^{-1}\mathcal{B}_{\mathcal{U}})$. This notation refers to the input factor of the measure-preserving lift, not to an optimized control entropy.

3. ENTROPY UNDER A COMMON RADIAL MOTION

Here h_{inv} denotes exact finite-catalogue invariance entropy, h_{out} uses ambient neighborhoods of the constraint, and h_{inv}^+ uses finite catalogues of viable reference-input pairs. Their precise quantifiers and limit orders are given in section 2. These conventions are fixed in the preceding section.

Let $A \in GL(d, \mathbb{R})$, let $B \subset \mathbb{R}^d$ be a compact convex body with $0 \in \text{int } B$, and let $U \subset \mathbb{R}^d$ be nonempty and compact. Assume that $\forall y \in B \exists v \in U : Ay + v \in B$. Let $R : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing smooth diffeomorphism with $R(0) = 0$ and $0 < R(s) < s$ for $0 < s \leq 1$. Write $\rho = R'(0) \in (0, 1]$, put $Q_B = \{(s, sy) : 0 \leq s \leq 1, y \in B\}$, and consider

$$(3.1) \quad f((s, z), v) = \left(R(s), \frac{R(s)}{s} (Az + sv) \right),$$

using the smooth extension $R(s)/s = \rho$ at $s = 0$. For $L \in GL(d, \mathbb{R})$, write

$$(3.2) \quad \chi^+(L) = \sum_{j=1}^d \max\{0, \log |\lambda_j(L)|\},$$

counting algebraic multiplicities. In the next theorem, superscripts f and g distinguish (3.1) from its normalized affine system $g(y, v) = Ay + v$.

Theorem 3.1 (Cone transference and spectral formulas). *Under the preceding assumptions:*

- (i) Q_B is a full-dimensional compact convex controlled invariant set, and every f_v is a global C^∞ diffeomorphism of $\mathbb{R}^{d+1}$. For every integer $n \geq 0$, including possible infinite values,

$$(3.3) \quad r_{\text{inv}}^f(n, Q_B, Q_B) = r_{\text{inv}}^g(n, B, B).$$

- (ii) For every compact $K \subset Q_B$ with $\text{vol}_{d+1}(K) > 0$,

$$(3.4) \quad h_{\text{out}}^f(K, Q_B) = h_{\text{inv}}^{f,+}(K, Q_B) = \chi^+(\rho A), \quad h_{\text{inv}}^f(K, Q_B) \geq \chi^+(A).$$

For each fixed $\varepsilon > 0$, the normalized logarithms defining the outer and tracking counts have actual limits, both equal to $\chi^+(\rho A)$.

- (iii) Suppose, additionally, that there is $\eta_0 > 0$ such that

$$(3.5) \quad \forall y \in B \quad \exists v \in U : \quad \text{dist}(Ay + v, \mathbb{R}^d \setminus B) \geq \eta_0.$$

Then every such K satisfies

$$(3.6) \quad \lim_{n \rightarrow \infty} \frac{1}{n} \log r_{\text{inv}}^f(n, K, Q_B) = h_{\text{inv}}^f(K, Q_B) = \chi^+(A).$$

Theorem 3.2 (Nonlinear perturbations preserving the entropy rates). *Let $q > 1$, let $U \subset \mathbb{R}$ be finite and nonempty, and suppose that for some $\eta > 0$*

$$(3.7) \quad \forall y \in [-1, 1] \quad \exists v \in U : |qy + v| \leq 1 - 2\eta.$$

Let R satisfy the radial assumptions above, put $\rho = R'(0)$, and extend $t(s) = R(s)/s$ smoothly at zero. For every $v \in U$, let $P_v : \mathbb{R}^2 \rightarrow \mathbb{R}$ be smooth, with

$$(3.8) \quad P_v(0, 0) = 0, \quad DP_v(0, 0) = 0, \quad \|P\|_{C_b^2} < \min\{\eta, q/2\},$$

where $\|P\|_{C_b^2}$ is the maximum, over $v \in U$, of the global suprema of $|P_v|$, the Euclidean operator norm of DP_v , and that of D^2P_v . Define

$$(3.9) \quad f_v(s, z) = (R(s), t(s)[qz + sv + P_v(s, z)]), \quad Q = \{(s, z) : 0 \leq s \leq 1, |z| \leq s\}.$$

Then each f_v is a global smooth diffeomorphism, Q is controlled invariant, and for every compact $K \subset Q$ of positive area,

$$(3.10) \quad \lim_{n \rightarrow \infty} \frac{1}{n} \log r_{\text{inv}}(n, K, Q) = h_{\text{inv}}(K, Q) = \log q,$$

$$(3.11) \quad h_{\text{out}}(K, Q) = h_{\text{inv}}^+(K, Q) = \max\{0, \log(\rho q)\}.$$

All feasible states at time n , uniformly over initial states and inputs, lie in $R^n(1)Q$. Thus every pair of feasible state trajectories is asymptotic.

The perturbations form a nonempty open ball relative to the space of bounded C^2 perturbations with zero first jet, intersected with the smooth functions. The radial map and the control alphabet are held fixed. This is a persistence statement in a specified infinite-dimensional class; it does not assert openness under arbitrary changes of the radial dynamics or of the linearization.

Corollary 3.3 (Equal positive control entropies with zero state entropy). *Set $R(s) = \text{arsinh } s$, $q = 2$, $U = \{-3/2, 0, 3/2\}$, and*

$$(3.12) \quad P_v(s, z) = e_v \tanh(s) \tanh(z), \quad \max_{v \in U} |e_v| < 1/64.$$

The maps (3.9) are global smooth diffeomorphisms, and for every compact $K \subset Q$ of positive area,

$$(3.13) \quad h_{\text{inv}}(K, Q) = h_{\text{out}}(K, Q) = h_{\text{inv}}^+(K, Q) = \log 2.$$

Every pair of feasible state trajectories has distance at most $2R^n(1) \rightarrow 0$ at time n , with $R^n(1) \sim \sqrt{3/n}$. The true constrained lift has total topological entropy $\log 3$, uniform entropy over input fibers zero, and state-trajectory entropy zero. Every invariant lift measure is concentrated on the common fixed state and has zero conditional Kolmogorov–Sinai entropy over the input factor.

The two-symbol unperturbed example and its exact finite-horizon counts appear in theorem 6.1 and corollary 6.3; the proof of corollary 3.3 is given in section 6.3. The equalities in this corollary show that the phenomenon persists when all three control entropies are positive. It is not explained by an exact-versus-outer entropy gap.

The proofs have three separate components. Linear Bowen coverings and unstable sectional volumes yield theorem 3.1, without diagonalizability or a hyperbolicity assumption. A finite-horizon reduction for tracking catalogues, proved in proposition 4.3, supplies the nonlinear upper bound in theorem 3.2; uniform small-radius blocks give its exact upper bound. Finally, theorem 7.1 uses a common Lyapunov inequality to identify the state factor, the input fibers, all invariant measures, and the eventual core of the true constrained lift. Positivity of control entropy is supplied by the spectral and perturbation theorems, not by the Lyapunov inequality alone. Corollary 6.4 also realizes every prescribed pair of finite rates $a > 0$, $0 \leq b \leq a$, as exact and outer invariance entropies.

4. COVERING ESTIMATES AND CONTROL-HORIZON REDUCTION

4.1. Linear coverings and radial estimates.

Lemma 4.1 (Linear Bowen coverings). *Let $L \in GL(d, \mathbb{R})$ and let $B \subset \mathbb{R}^d$ be compact. For each $\delta > 0$, there are finite sets $E_n \subset B$ such that every $y \in B$ has $c \in E_n$ satisfying*

$$\max_{0 \leq k \leq n} \|L^k(y - c)\| < \delta, \quad |E_n| \leq C_\delta(n+1)^{d(d-1)} \exp(n\chi^+(L)).$$

The constant may depend on L, B, δ , but not on n .

Proof. Work in real Jordan coordinates and group the real generalized eigenspaces according to eigenvalue modulus $r > 0$; a nonreal conjugate pair has its real dimension. On a group of dimension d_r , the Jordan formula and equivalence of norms give

$$\max_{0 \leq k \leq n} \|L^k w\| \leq C(n+1)^{d-1} \max\{1, r^n\} \|w\|.$$

Cover a fixed coordinate box containing B by half-open rectangular cells. In each coordinate of the group of modulus r , take side length $\delta/[C'(n+1)^{d-1} \max\{1, r^n\}]$, where C' is large enough that two points in a cell have Bowen distance less than δ . Choose one point of B in each cell meeting B . The number of cells is at most

$$C_\delta(n+1)^{d(d-1)} \prod_r \max\{1, r^n\}^{d_r}.$$

The product is $\exp(n\chi^+(L))$. This proves the assertion, including nontrivial Jordan blocks and unit-modulus eigenvalues. $\square$

Lemma 4.2 (Radial decay). *Suppose $R : \mathbb{R} \rightarrow \mathbb{R}$ is an increasing C^∞ diffeomorphism, $R(0) = 0$, and $0 < R(s) < s$ for $0 < s \leq 1$. Write $\rho = R'(0)$, so $0 < \rho \leq 1$. Then $R^n(1) \downarrow 0$, uniformly bounding $R^n(s)$ for $s \in [0, 1]$. For every $0 < \eta < \rho$ there is $C_\eta < \infty$ such that*

$$(4.1) \quad 0 < \frac{R^n(s)}{s} \leq C_\eta(\rho + \eta)^n \quad (0 < s \leq 1, n \geq 0).$$

For every $s_0 \in (0, 1]$ there is $c_{\eta, s_0} > 0$ such that

$$(4.2) \quad \frac{R^n(s)}{s} \geq c_{\eta, s_0}(\rho - \eta)^n \quad (s_0 \leq s \leq 1, n \geq 0).$$

The family $\{R^n|_{[0,1]} : n \geq 0\}$ is equicontinuous.

Proof. The limit of the decreasing sequence $R^n(1)$ is a fixed point of R in $[0, 1]$, hence is zero. Monotonicity gives $0 \leq R^n(s) \leq R^n(1)$. Choose $a > 0$ such that $\rho - \eta \leq R(t)/t \leq \rho + \eta$ for $0 < t \leq a$, and then N with $R^N(1) \leq a$. In the product

$$\frac{R^n(s)}{s} = \prod_{j=0}^{n-1} \frac{R(R^j(s))}{R^j(s)},$$

all factors with $j \geq N$ have the indicated bounds. The finitely many initial products have a uniform upper bound for $0 < s \leq 1$, using their continuous extensions at zero. They have a positive minimum on $[s_0, 1]$. Absorbing these finite products proves (4.1) and (4.2), also for $n < N$.

For equicontinuity, given $\varepsilon > 0$, choose N with $R^N(1) < \varepsilon$. All maps with index at least N have range in $[0, \varepsilon)$. Uniform continuity of the finitely many preceding maps supplies a single initial-distance bound for the remaining indices. $\square$

4.2. Finite-horizon catalogues under uniform convergence. The following estimate will allow the tracking upper bound to survive nonlinear perturbations. It uses exact controls for only a fraction of the requested tracking horizon.

Proposition 4.3 (Shortening the exact-control horizon). *Let X be a normed vector space, let U be compact, let $f : X \times U \rightarrow X$ be continuous, and let $Q \subset X$ be compact and controlled invariant. Suppose that $p \in Q$ and*

$$(4.3) \quad b_n := \sup\{\|\varphi(n, x, u) - p\| : (u, x) \in \Lambda_Q\} \rightarrow 0.$$

If $H = h_{\text{inv}}(Q, Q) < \infty$, then

$$(4.4) \quad h_{\text{out}}(Q, Q) \leq h_{\text{inv}}^+(Q, Q) \leq H.$$

Suppose in addition that $b_n \leq Ca^n$ for constants $C > 0$, $0 < a < 1$, and that all maps f_v are L -Lipschitz on the same ball about p , with $L > 1$. Then

$$(4.5) \quad h_{\text{inv}}^+(Q, Q) \leq \frac{\log L}{\log(L/a)} H.$$

The same upper bounds hold after restricting the initial set to any compact subset of Q .

Proof. Fix a sufficiently small $\varepsilon > 0$. Choose N so that $2b_k < \varepsilon/2$ for every $k \geq N$. Uniform continuity of the finitely many solution maps on $Q \times U^N$ gives $\delta > 0$ such that two initial states at distance less than δ , subjected to the same input prefix, have distance less than $\varepsilon/2$ at times $0, \dots, N$. Cover Q by C_ε sets of diameter less than δ .

Take an exact spanning catalogue of horizon $m \geq N$. For each catalogue word and each covering set that contains an initial state feasible for this word through time m , choose one such state as a reference. Extend its word to an infinite feasible input by controlled invariance. The extension may depend on the chosen reference; the number of reference-input pairs is at most

$$(4.6) \quad C_\varepsilon r_{\text{inv}}(m, Q, Q).$$

Every initial state can be assigned a pair sharing its original catalogue word and its covering set. The actual and reference trajectories have distance less than $\varepsilon/2$ through time N . From time N through time m , both trajectories are feasible, and their distance is at most $2b_k < \varepsilon/2$. Here the bound for infinite feasible trajectories applies to each finite

feasible prefix because that prefix can be continued in Q . Taking $m = n$, then normalized logarithms and the prescribed limits, proves (4.4).

For the sharper estimate let the common Lipschitz ball have radius r . Decrease ε so that $\varepsilon < r/4$, and increase N so that $b_k < r/4$ for $k \geq N$. Put

$$(4.7) \quad \theta = \frac{\log L}{\log L - \log a} \in (0, 1), \quad m = \lceil \theta n + c_\varepsilon \rceil,$$

where $c_\varepsilon > 0$ is chosen so that $2C \exp[-c_\varepsilon \log(L/a)] < \varepsilon/2$. For large n , $N \leq m \leq n$. Use the refined horizon- m catalogue just constructed. At time m , the two assigned trajectories have distance at most $2Ca^m$. For $m \leq k \leq n$, iteration of the Lipschitz bound gives

$$(4.8) \quad \|\varphi(k, x, u) - \varphi(k, a_i, u)\| \leq 2Ca^m L^{k-m} \leq 2Ca^m L^{n-m} < \varepsilon/2.$$

This iteration is justified inductively: the reference remains within $r/4$ of p , and the asserted error puts the actual state inside the ball of radius $r/2$; hence both states lie in the common Lipschitz domain at the next step. The last inequality follows from (4.7), since $n \log L - m \log(L/a) \leq -c_\varepsilon \log(L/a)$. Consequently

$$(4.9) \quad r_{\text{inv}}^+(n, \varepsilon, Q, Q) \leq C_\varepsilon r_{\text{inv}}(\lceil \theta n + c_\varepsilon \rceil, Q, Q).$$

Because $H < \infty$, the last spanning number is finite for all sufficiently large horizons. Dividing its logarithm by n , using $m/n \rightarrow \theta$, and taking the upper limit gives the bound θH . Finally let $\varepsilon \downarrow 0$. Restriction of the initial set only decreases all the counts. $\square$

5. ENTROPY FORMULAS ON A SOLID CONE

For this section fix the following assumptions:

- (i) $A \in GL(d, \mathbb{R})$, $B \subset \mathbb{R}^d$ is compact and convex, and $0 \in \text{int } B$;
- (ii) $U \subset \mathbb{R}^d$ is nonempty and compact, and

$$(5.1) \quad \forall y \in B \quad \exists v \in U : Ay + v \in B;$$

- (iii) R satisfies the assumptions of lemma 4.2, and $\rho = R'(0)$.

Write

$$t(s) = \begin{cases} R(s)/s, & s \neq 0, \\ \rho, & s = 0, \end{cases} \quad Q_B = \{(s, sy) : 0 \leq s \leq 1, y \in B\}, \quad p = (0, 0).$$

We use Euclidean distances in the ambient space $\mathbb{R}^{d+1}$. The controlled maps are

$$(5.2) \quad f((s, z), v) = (R(s), t(s)(Az + sv)), \quad v \in U.$$

The normalized affine system is $g(y, v) = Ay + v$; superscripts f, g distinguish their spanning numbers when needed.

The statement of theorem 3.1 is proved in the next four subsections.

The positive-volume hypothesis in part (ii) is a lower-bound hypothesis, not a controllability condition. The upper estimates hold even on the whole cone. Condition (3.5) is used only for the exact upper bound. It is sufficient and is not asserted to be necessary; the finite digit systems below attain that bound without satisfying this condition.

5.1. Smoothness, normalization, and exact transference.

Proof of theorem 3.1(i). The identity $t(s) = \int_0^1 R'(\theta s) d\theta$ shows that t is smooth and strictly positive on $\mathbb{R}$. For fixed v , the inverse of f_v at (s', z') is

$$s = R^{-1}(s'), \quad z = A^{-1}(z'/t(s) - sv).$$

This is globally defined and smooth. The set Q_B is the convex hull of p and $\{1\} \times B$, so is compact and convex; since B has interior, this hull has nonempty interior in $\mathbb{R}^{d+1}$.

For an initial state (s, sy) with $s > 0$, let $s_k = R^k(s)$. Direct substitution gives

$$(5.3) \quad \varphi(k, (s, sy), u) = (s_k, s_k y_k), \quad y_k = A^k y + b_k(u), \quad b_k(u) = \sum_{j=0}^{k-1} A^{k-1-j} u_j.$$

Here $b_0 = 0$. Thus this state trajectory belongs to Q_B exactly when $y_k \in B$. Recursive use of (5.1) gives an infinite viable normalized control from every $y \in B$. The apex is fixed under every input.

Any finite catalogue spanning B for g therefore spans every positive radial slice and the apex for f . Conversely, any catalogue spanning Q_B must span the slice $\{(1, y) : y \in B\}$; normalization shows that precisely the same catalogue spans B for g . Existence of a finite catalogue in one problem is equivalent to its existence in the other, and minimizing proves (3.3). $\square$

5.2. Finite outer and tracking catalogues.

Upper bounds in theorem 3.1(ii). Fix $\varepsilon > 0$ and $0 < \eta < \rho$, and put $\alpha = \rho + \eta$. Apply lemma 4.1 to $L = \alpha A$ and a radius $\delta > 0$ to be specified. For each of the finitely many centers $c \in E_n \subset B$, choose once and for all an infinite normalized control u^c feasible from c . Denote its normalized trajectory by $c_k \in B$.

For $y \in B$, choose $c \in E_n$ with Bowen distance less than δ . The trajectories from (s, sy) and (s, sc) under the same u^c differ by

$$(0, R^k(s)A^k(y - c)).$$

By (4.1), $s \leq 1$, and the covering property, its norm is less than $C_\eta \delta$ for $0 \leq k \leq n$. Choose $\delta < \varepsilon/(2C_\eta)$. The reference trajectory from (s, sc) is in Q_B , so the actual trajectory is in Q_B^ε . The same single control u^c works for all radial values associated with that center; the spanning catalogue is finite. Consequently

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log r_{\text{inv}}^f(n, Q_B, Q_B^\varepsilon) \leq \chi^+((\rho + \eta)A).$$

For tracking, radial reference states must also belong to a *finite* catalogue. Put $C_B = \max_{y \in B} \|(1, y)\|$. Equicontinuity in lemma 4.2 and compactness of $[0, 1]$ give a finite set $T \subset [0, 1]$, independent of n , such that every $s \in [0, 1]$ has $t \in T$ satisfying

$$\sup_{k \geq 0} |R^k(s) - R^k(t)| < \varepsilon/(2C_B).$$

Use the finite pairs $((t, tc), u^c)$, $c \in E_n$, $t \in T$. With $t_k = R^k(t)$, their trajectory difference from (s, sy) is

$$(s_k - t_k, (s_k - t_k)c_k + s_k A^k(y - c)).$$

Its norm is bounded by $C_B |s_k - t_k| + C_\eta \delta < \varepsilon$. Every reference trajectory is feasible, including those with $t = 0$. Multiplication by the fixed number $|T|$ does not change the exponential rate. Hence the tracking upper rate is also at most $\chi^+((\rho + \eta)A)$, for every fixed $\varepsilon > 0$. Letting $\eta \downarrow 0$ gives the desired upper bound $\chi^+(\rho A)$; the finite sum defining χ^+ is continuous under this scalar variation. Restricting the initial set to K preserves both bounds. $\square$

5.3. Unstable-volume lower bounds.

Lower bounds in theorem 3.1(ii). There exists $s_0 > 0$ such that

$$K_0 = K \cap \{(s, z) : s \geq s_0\}$$

has positive $(d + 1)$ -dimensional volume: the increasing union of these truncated sets as $s_0 \downarrow 0$ differs from K by a subset of $\{s = 0\}$, which has zero ambient volume.

Let E_+ be the real sum of the generalized eigenspaces of A for which $\rho|\lambda| > 1$. Let E_- be the invariant complementary sum, let $r = \dim E_+$, and write $z = z_+ + z_-$. If $r = 0$, the claimed outer lower bound is simply nonnegativity. Suppose $r > 0$, and write $A_+ = A|_{E_+}$.

Fix a control u . For arbitrary initial z , (5.3) gives the ambient last coordinate

$$(5.4) \quad z_n = a_n(s)A^n z + R^n(s)b_n(u), \quad a_n(s) = R^n(s)/s.$$

Let $W_n(u) \subset K_0$ consist of initial points whose trajectory stays in Q_B^ε at times $0, \dots, n$. Since Q_B^ε is bounded, its final E_+ -coordinate belongs to a fixed ball, with radius depending on ε but not on n, u . For each fixed s, z_- , the map from z_+ to that coordinate is an affine map with linear part $a_n(s)A_+^n$. The r -dimensional volume of the corresponding section of $W_n(u)$ is therefore at most

$$C_\varepsilon a_n(s)^{-r} |\det A_+|^{-n} \leq C_{\varepsilon, \eta, s_0} ((\rho - \eta)^r |\det A_+|)^{-n},$$

where (4.2) was used. This bound is unaffected by the translation, which may depend on u, s .

The s, z_- projection of K_0 is contained in a fixed bounded box. The change to the chosen linear coordinates has a constant nonzero Jacobian. Fubini's theorem consequently gives

$$\text{vol}_{d+1}(W_n(u)) \leq C'_{\varepsilon, \eta, K} ((\rho - \eta)^r |\det A_+|)^{-n}.$$

The sets are measurable, since finite-time trajectories are continuous. If M controls span K , their sets $W_n(u)$ cover K_0 . Thus

$$M \geq \frac{\text{vol}_{d+1}(K_0)}{C'_{\varepsilon, \eta, K}} ((\rho - \eta)^r |\det A_+|)^n.$$

Taking the lower limit of normalized logarithms and then letting $\eta \downarrow 0$ yields

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log r_{\text{inv}}^f(n, K, Q_B^\varepsilon) \geq r \log \rho + \log |\det A_+| = \chi^+(\rho A).$$

Together with the upper bound this proves an actual time limit for every $\varepsilon > 0$. Inequality (2.4) and the tracking upper bound give the same conclusion for tracking counts. The subsequent limit $\varepsilon \downarrow 0$ has the stated value.

For exact containment use instead the unstable subspace of A , with $|\lambda| > 1$. From $z_n/R^n(s) \in B$ we obtain

$$A^n z + s b_n(u) \in s B \subset B.$$

For fixed s, z_- , the relevant linear part on the unstable subspace is now A_+^n , without a radial factor. The same sectional-volume and Fubini argument gives

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log r_{\text{inv}}^f(n, K, Q_B) \geq \chi^+(A).$$

This statement also covers infinite exact spanning numbers. It finishes part (ii). $\square$

5.4. The exact upper bound with an interior margin.

Proof of theorem 3.1(iii). Choose $\delta < \eta_0$ and a Bowen covering $E_n \subset B$ for the matrix A from lemma 4.1. At each center c , recursively use (3.5) to choose an infinite normalized control with

$$\text{dist}(c_k, \mathbb{R}^d \setminus B) \geq \eta_0 \quad \text{for every } k \geq 1.$$

For any y assigned to c , its normalized trajectory under that same control satisfies $\|y_k - c_k\| = \|A^k(y - c)\| < \delta$ for $0 \leq k \leq n$. It follows that $y_k \in B$ for $k \geq 1$, and $y_0 \in B$ already holds. Thus these finitely many controls span B , and by exact transference they span Q_B . Their number has upper exponential rate at most $\chi^+(A)$. The exact lower bound just proved gives (3.6). $\square$

Remark 5.1 (Where the assumptions enter). Convexity and $0 \in \text{int } B$ make the constraint a solid convex cone; the proof uses boundedness for outer estimates and positive volume for lower estimates. No diagonalizability or hyperbolicity of A is needed. The smooth radial hypotheses ensure global invertibility, the estimates in lemma 4.2, and uniform decay. Condition (3.5) is automatic if $U \supset -AB$, since each normalized state can be reset to zero. It also holds for appropriate finite redundant alphabets. The theorem does not assert that an arbitrary viable compact affine system has finite exact invariance entropy without this condition.

Remark 5.2 (Why a Lie-group formula is not a direct substitute). In the author version of [13, Definition 1], a discrete-time linear system on a Lie group has the form $f_v(g) = a_v f_0(g)$, with f_0 an automorphism. If all such maps fix one point p , then $a_v = p f_0(p)^{-1}$ is independent of v , and all controlled maps coincide. The cone maps have a common fixed apex and are distinct for distinct v . Thus they do not have this representation on any group structure that represents the same maps on the whole state space. The normalization $y = z/s$ is defined only for $s > 0$; deleting the apex removes compactness of the viable constraint and does not justify transferring a compact-constraint entropy formula. This observation excludes that direct application, not every possible use of homogeneous-space theory.

5.5. Nonlinear perturbations with fixed first jet. We prove theorem 3.2. Exact transference no longer holds for these systems: the normalized dynamics depends on the current radius. Its entropy can nevertheless be computed by combining finite blocks at small radii with one-dimensional sectional-volume estimates.

Proof of theorem 3.2. Choose δ with $\|P\|_{C_b^2} < \delta < \min\{\eta, q/2\}$. Taylor's formula at the origin gives, for $0 < s \leq 1$ and $|y| \leq 1$,

$$(5.5) \quad |P_v(s, sy)| \leq \frac{1}{2} \delta s^2 (1 + y^2) \leq \delta s^2, \quad |\partial_z P_v(s, z)| \leq \delta \sqrt{s^2 + z^2}.$$

The second inequality holds for all (s, z) ; the global bound $|\partial_z P_v| < \delta$ also holds.

Smoothness and viability. The first coordinate determines s uniquely from the image. For this s , the map $z \mapsto qz + sv + P_v(s, z)$ has derivative at least $q - \delta > 0$, tends to the two respective infinities at the ends of the real line, and is therefore a bijection. The implicit function theorem gives a smooth inverse depending on s and the image coordinate. Positivity and smoothness of t , already established above, prove global smooth invertibility.

At positive radius the normalized coordinate satisfies

$$(5.6) \quad y_{k+1} = qy_k + u_k + \frac{P_{u_k}(s_k, s_k y_k)}{s_k}, \quad s_k = R^k(s).$$

At any $|y_k| \leq 1$, choose u_k by (3.7). The perturbation term has absolute value at most $\delta s_k \leq \delta < \eta$, so $|y_{k+1}| \leq 1 - \eta$. The apex is fixed under every input. This proves controlled invariance, with a uniform normalized interior margin. Every finite feasible trajectory satisfies $0 \leq s_k \leq R^k(1)$ and $|z_k| \leq s_k$. Hence all its time- k states lie in $R^k(1)Q$.

Exact upper bound. Fix an integer block length $N \geq 1$. Cover $[-1, 1]$ by intervals of radius $\gamma_N = (\eta/4)q^{-N}$, with centers in $[-1, 1]$, using at most $C_\eta q^N$ intervals, where C_η is independent of N . In the unperturbed affine system $y^+ = qy + v$, choose from each center a length- N word whose reference states at positive times all satisfy $|c_k| \leq 1 - 2\eta$. This is possible by recursive use of (3.7).

Choose $a_N \in (0, 1]$ so small that

$$(5.7) \quad \delta a_N \sum_{j=0}^{N-1} q^j \leq \eta/4.$$

For initial radius $0 < s \leq a_N$ and an initial ratio y assigned to a center, apply its unperturbed word to the perturbed system. As long as the perturbed ratios remain in $[-1, 1]$, their errors relative to the unperturbed reference obey

$$(5.8) \quad |y_k - c_k| \leq q^k \gamma_N + \delta a_N \sum_{j=0}^{k-1} q^j \leq \eta/2 \quad (0 \leq k \leq N).$$

This estimate proves its own feasibility premise by induction: the reference at each positive time has margin 2η , while the error is at most $\eta/2$. Thus these same $C_\eta q^N$ words work simultaneously for all radii at most a_N . No radius-dependent infinite catalogue has been introduced.

Choose N_0 with $R^{N_0}(1) \leq a_N$. All words in U^{N_0} form a finite exact catalogue for this transient, by controlled invariance. Thereafter concatenate arbitrary words from the small-radius block catalogue. Each current radius is still at most a_N , so the argument can be applied successively to every block. For $n \geq N_0$, using a prefix of the final block if necessary, this gives

$$(5.9) \quad r_{\text{inv}}(n, Q, Q) \leq |U|^{N_0} (C_\eta q^N)^{\lceil (n-N_0)/N \rceil}.$$

Consequently the upper entropy is at most $\log q + N^{-1} \log C_\eta$. Letting $N \rightarrow \infty$ proves the exact upper bound $\log q$.

Exact lower bound. For fixed $s > 0$ and a fixed input word u , let $I_n(s, u)$ be the set of $z \in [-s, s]$ whose states satisfy Q through time n . This set is an interval, possibly empty or a singleton: every vertical solution map is strictly increasing, and $I_n(s, u)$ is the intersection of inverse images of vertical intervals. In particular every point between two members has a feasible trajectory through time n .

Write $Z_n(s, z, u)$ for the second state coordinate. Along this interval the chain rule yields

$$(5.10) \quad \partial_z Z_n(s, z, u) = \frac{R^n(s)}{s} \prod_{j=0}^{n-1} [q + \partial_z P_{u_j}(s_j, z_j)].$$

The product of the factors $t(s_j)$ telescopes to $R^n(s)/s$. For any $\zeta \in (0, q-1)$, uniform feasible convergence and (5.5) imply that every factor after a fixed time N_ζ is at least $q-\zeta$, independently of s, z, u . The finitely many preceding factors are at least $q-\delta$. There is therefore a constant $c_\zeta > 0$, independent of the word and of n, s , such that

$$(5.11) \quad \partial_z Z_n(s, z, u) \geq c_\zeta \frac{R^n(s)}{s} (q-\zeta)^n \quad \text{on } I_n(s, u).$$

The image interval at time n has length at most $2R^n(s)$. Integrating the derivative between any two points of $I_n(s, u)$ proves

$$(5.12) \quad |I_n(s, u)| \leq \frac{2s}{c_\zeta} (q-\zeta)^{-n}.$$

The feasible set of one word is closed and hence measurable. Fubini's theorem, applied over $0 < s \leq 1$, bounds its area in Q by a constant times $(q-\zeta)^{-n}$. The slice $s = 0$ has zero area. Covering a compact $K \subset Q$ of positive area consequently requires at least $c_K (q-\zeta)^n$ words. Taking the lower time limit and then $\zeta \downarrow 0$, and combining with the upper bound, proves the first line of (3.10), including existence of its time limit.

Outer lower bound. Choose $s_0 > 0$ so that $K_0 = K \cap \{s \geq s_0\}$ has positive area. Fix small $\varepsilon > 0$. Replace $I_n(s, u)$ by the initial interval of points in $[-s, s]$ whose trajectory remains in Q^ε through time n . This is still an interval: Q^ε is convex, all its vertical sections are intervals, and all vertical solution maps are increasing. For an outer-feasible

state with radial coordinate s_j , comparison with a point $(r, w) \in Q$ at distance less than ε gives

$$(5.13) \quad |z_j| < s_j + 2\varepsilon.$$

After a time depending only on ε , $s_j \leq R^j(1) \leq \varepsilon$, and hence $|z_j| \leq 3\varepsilon$. By (5.5), the late factors in (5.10) are bounded below by $q - \omega_\varepsilon$, where $\omega_\varepsilon = \delta\sqrt{10}\varepsilon \rightarrow 0$. Take ε small enough that this number is positive. Absorbing the finitely many earlier factors into $c_\varepsilon > 0$, the derivative on the entire outer-feasible initial interval is at least

$$(5.14) \quad c_\varepsilon \frac{R^n(s)}{s} (q - \omega_\varepsilon)^n.$$

Its time- n image has length at most $2(1 + 2\varepsilon)$. For $s \geq s_0$, the radial lower bound in lemma 4.2 then bounds its initial length by

$$(5.15) \quad C_{\varepsilon, \xi, s_0} [(\rho - \xi)(q - \omega_\varepsilon)]^{-n}, \quad 0 < \xi < \rho.$$

The mean-value estimate remains valid for open or half-open intervals by applying it between any two of their points. Fubini on K_0 , followed by the covering argument, gives a lower time rate at least $\log[(\rho - \xi)(q - \omega_\varepsilon)]$. Nonnegativity supplies the maximum with zero. Let $\xi \downarrow 0$ and then $\varepsilon \downarrow 0$, in that order after the time limit, to obtain

$$(5.16) \quad h_{\text{out}}(K, Q) \geq \max\{0, \log(\rho q)\}.$$

Tracking upper bound. If $\rho = 1$, uniform feasible convergence and proposition 4.3 give $h_{\text{inv}}^+(Q, Q) \leq h_{\text{inv}}(Q, Q) = \log q$, which matches (5.16). Suppose $0 < \rho < 1$. For any $a \in (\rho, 1)$, lemma 4.2 and feasibility give a uniform bound $b_n \leq C_a a^n$. At the apex,

$$(5.17) \quad Df_v(p) = \begin{pmatrix} \rho & 0 \\ \rho v & \rho q \end{pmatrix}.$$

For $M > 0$, equip $\mathbb{R}^2$ with the equivalent norm $\|(s, z)\|_M = \max\{M|s|, |z|\}$. The corresponding operator norm of (5.17) is at most $\rho q + \rho \max_{v \in U} |v|/M$. For any $L > \max\{1, \rho q\}$, choose M large and then a common convex neighborhood of p small enough that all f_v are L -Lipschitz there. Finiteness of U makes the neighborhood uniform. Equivalent norms leave these entropy definitions unchanged, since each metric tolerance can be bounded by a constant multiple of the other.

Applying proposition 4.3 in this norm yields

$$(5.18) \quad h_{\text{inv}}^+(Q, Q) \leq \log q \frac{\log L}{\log L - \log a}.$$

Let $a \downarrow \rho$ and $L \downarrow \max\{1, \rho q\}$. If $\rho q > 1$, the right-hand side tends to $\log(\rho q)$; if $\rho q \leq 1$, it tends to zero. This includes the critical case $\rho q = 1$. Restricting the initial set to K and using $h_{\text{out}} \leq h_{\text{inv}}^+$ completes the proof. $\square$

Remark 5.3 (What the nonlinear calculation does and does not preserve). The exact entropy is determined by the limiting normalized expansion q . Its upper bound uses finite blocks that work uniformly for a whole radial interval, whereas its lower bound uses derivatives on vertical intervals. The outer lower bound additionally uses the smallness of the derivative perturbation near the apex; the tracking upper bound uses proposition 4.3. Exact finite-horizon transference and the digit tiling are not claimed for the perturbed system. Nor do we assert a common fixed- ε time limit for its outer or tracking counts. The entropy identities use exactly the iterated limits in section 2.

6. FINITE ALPHABETS AND EQUAL POSITIVE ENTROPIES

6.1. Exact counts in arbitrary dimension. For integers $q_1, \dots, q_d \geq 2$, set

$$U_{q_i} = \{q_i - 1 - 2j : 0 \leq j < q_i\}, \quad A = \text{diag}(q_1, \dots, q_d), \quad U = \prod_{i=1}^d U_{q_i}, \quad M = \prod_{i=1}^d q_i.$$

Take $B = [-1, 1]^d$, $Q = Q_B$, and

$$(6.1) \quad K_d = [1/2, 3/4] \times [-1/4, 1/4]^d.$$

This compact set is contained in $\text{int } Q$ and has positive ambient volume. In particular, $|z_i| \leq 1/4 < 1/2 \leq s < 1$ on K_d .

Theorem 6.1 (Finite digit family). *For every radial map R satisfying lemma 4.2, the preceding finite-alphabet cone system satisfies, for all integers $n \geq 0$,*

$$(6.2) \quad r_{\text{inv}}(n, Q, Q) = M^n, \quad 2^{-d}M^n \leq r_{\text{inv}}(n, K_d, Q) \leq M^n.$$

Consequently

$$h_{\text{inv}}(K_d, Q) = h_{\text{inv}}(Q, Q) = \sum_{i=1}^d \log q_i,$$

and

$$(6.3) \quad h_{\text{out}}(K_d, Q) = h_{\text{out}}(Q, Q) = h_{\text{inv}}^+(K_d, Q) = h_{\text{inv}}^+(Q, Q) = \sum_{i=1}^d \max\{0, \log(\rho q_i)\}.$$

For any two initial states $x, y \in Q$ and any two controls u, v respectively feasible for all nonnegative times,

$$(6.4) \quad \|\varphi(n, x, u) - \varphi(n, y, v)\| \leq 2\sqrt{d} R^n(1) \rightarrow 0.$$

All individual controlled maps are global smooth diffeomorphisms.

Proof. For one coordinate with expansion q , the intervals

$$I_v = \left[\frac{-1-v}{q}, \frac{1-v}{q} \right], \quad v \in U_q,$$

tile $[-1, 1]$ with disjoint interiors. The map $y \mapsto qy + v$ sends I_v onto $[-1, 1]$, and its inverse branch sends all of $[-1, 1]$ into I_v . Hence every point admits an infinite viable sequence, and every word of length n has a viable interval of length $2q^{-n}$. More explicitly, composing the inverse branches maps $[-1, 1]$ to the set viable for that word; all intermediate inverse images remain in $[-1, 1]$.

Products give one viable box for each length- n word over U , of volume $2^d M^{-n}$. The M^n words span the entire cube B . Conversely, for any fixed word the final-time condition alone confines its viable initial points to an affine preimage of B of volume $2^d M^{-n}$. Covering B , of volume 2^d , requires at least M^n words. Transference proves the first identity in (6.2).

To obtain the lower bound for K_d , restrict any spanning family to the slice $s = 1/2$. Its normalized initial coordinates contain exactly $[-1/2, 1/2]^d$, of d -volume 1. One control word covers at most $2^d M^{-n}$ of that slice in normalized coordinates. At least $2^{-d} M^n$ words are required. The upper bound follows by restricting a catalogue for Q . This gives the exact entropy formula without an interior margin assumption. Formula (6.3) follows from theorem 3.1(ii).

Every feasible state at time n belongs to $\{(s, z) : 0 \leq s \leq R^n(1), |z_i| \leq s\} = R^n(1)Q$. The diameter of Q is $2\sqrt{d}$: its vertices are the apex and the vertices $(1, \pm 1, \dots, \pm 1)$, and the largest distance between them is $2\sqrt{d}$. A convex hull has the same diameter as the supremum of the distances between its generating vertices. This proves (6.4). Smooth invertibility was proved in theorem 3.1(i). $\square$

Remark 6.2 (The planar linear-radial system). With $d = 1$ and $R(s) = \rho s$, $0 < \rho < 1$, the system is

$$f((s, z), v) = (\rho s, \rho(qz + sv)), \quad v \in U_q, \quad Q = \{0 \leq s \leq 1, |z| \leq s\}.$$

Its fixed-input maps are linear isomorphisms with determinant $\rho^2 q$; its dependence on state and input is bilinear. The counts reduce to

$$r_{\text{inv}}(n, Q, Q) = q^n, \quad q^n/2 \leq r_{\text{inv}}(n, K_1, Q) \leq q^n.$$

The exact rate is $\log q$, the outer and tracking rates are $\max\{0, \log(q\rho)\}$, and the distance of any two feasible trajectories is at most $2\rho^n$. In particular both control entropies are positive for $1/q < \rho < 1$.

6.2. Subexponential radial decay removes the entropy gap.

Corollary 6.3 (Equal positive entropies). *Let $R(s) = \text{arsinh } s$, $d = 1$, and $q = 2$, with $U_2 = \{-1, 1\}$. On the same solid triangle Q and interior rectangle K_1 , every controlled map is a global smooth diffeomorphism and*

$$(6.5) \quad h_{\text{inv}}(K_1, Q) = h_{\text{out}}(K_1, Q) = h_{\text{inv}}^+(K_1, Q) = \log 2.$$

The same equalities hold for K_1 replaced by Q . All pairs of feasible state trajectories are asymptotic. Moreover,

$$R^n(1) \sim \sqrt{3/n}.$$

Proof. The function $R = \text{arsinh}$ is a smooth increasing bijection of $\mathbb{R}$ with smooth inverse $\sinh$, satisfies $0 < R(s) < s$ for $s > 0$, and has $R'(0) = 1$. Thus theorem 6.1 applies with $\rho = 1$, proving (6.5) and asymptoticity.

For the rate, write $s_n = R^n(1)$. Since $R(s) = s - s^3/6 + O(s^5)$ at zero, and $s_n \downarrow 0$,

$$s_{n+1}^{-2} - s_n^{-2} = s_n^{-2}[(1 - s_n^2/6 + O(s_n^4))^{-2} - 1] \rightarrow 1/3.$$

Telescoping and taking Cesàro averages gives $s_n^{-2}/n \rightarrow 1/3$, which proves the asserted asymptotic formula. $\square$

The equality case shows that a discrepancy between exact and neighborhood constraints is not responsible for the failure of a state Li–Yorke conclusion. A fixed positive ambient tolerance still requires exponentially many control signals. Nevertheless every signal that is infinitely feasible produces a state converging to the apex.

6.3. A nonlinear equality family.

Proof of corollary 3.3. For $q = 2$ and $U = \{-3/2, 0, 3/2\}$, the intervals of points for which one of $|2y + v|$ is at most $3/4$ cover $[-1, 1]$. Equivalently, the nearest of the three centers $-3/4, 0, 3/4$ is at distance at most $3/8$ from every $y \in [-1, 1]$. Thus (3.7) holds with $\eta = 1/8$.

The function $\tanh(s) \tanh(z)$ and its first two derivatives are bounded; its value and first derivative at the origin vanish. The absolute values of its two pure second derivatives are at most 2, and that of its mixed derivative is at most 1, so its Hessian operator norm is at most 3. In particular the bound $4 \max_v |e_v| < 1/16 < \eta$ dominates the required C_b^2 norm. Theorem 3.2 therefore gives the three entropy values and uniform feasible convergence. The diameter of Q is 2, giving the pair-distance bound. The radial asymptotic is proved in corollary 6.3.

On Q , the function $V(s, z) = s$ vanishes only at the apex and satisfies $V(f_v(s, z)) = R(V(s, z))$ for every feasible step. Apply theorem 7.1 with $\psi = R|_{[0,1]}$. Since $|U| = 3$, its lift, fiber, state-factor, and measure conclusions give all remaining assertions. $\square$

Taking at least one $e_v \neq 0$ makes the vertical controlled map nonlinear. More generally the entire perturbation ball in theorem 3.2, with this fixed alphabet and $R = \text{arsinh}$, has the same entropy equalities and the same Lyapunov conclusions. Redundancy of the alphabet supplies the interior margin that allows this perturbation argument; the minimal digit alphabet is used for exact counting, not for this persistence statement.

6.4. Prescribing the two control entropy rates.

Corollary 6.4 (Arbitrary ordered rates). *For every $a > 0$ and $b \in [0, a]$, there is a planar system of the form (5.2) with a finite input alphabet, global smooth invertible controlled maps, the triangle Q , and the rectangle K_1 , such that*

$$h_{\text{inv}}(K_1, Q) = h_{\text{inv}}(Q, Q) = a, \quad h_{\text{out}}(K_1, Q) = h_{\text{out}}(Q, Q) = b.$$

Its tracking entropies also equal b , and all pairs of feasible state trajectories are asymptotic.

Proof. Put $L = e^a > 1$, $m = \lceil L \rceil + 2$, and

$$c_j = -1 + \frac{2j}{m-1}, \quad U = \{-Lc_j : 0 \leq j < m\}, \quad A = L, \quad B = [-1, 1].$$

For every $y \in B$, a nearest center satisfies $|y - c_j| \leq 1/(m-1)$, so

$$|Ly - Lc_j| \leq \frac{L}{m-1} < 1.$$

Thus (3.5) holds with $\eta_0 = 1 - L/(m-1) > 0$. Exact entropy equals $\log L = a$ by theorem 3.1(iii). If $b < a$, choose $R(s) = e^{b-a}s$. Then $\rho L = e^b$ and the outer and tracking formula is b , including the threshold case $b = 0$. If $b = a$, choose $R(s) = \text{arsinh } s$; then $\rho = 1$ gives the same conclusion. Uniform radial decay and feasibility place all trajectories in $R^n(1)Q$, proving asymptoticity. $\square$

This realizes the full range of ordered finite values in dimension two, with a finite alphabet depending on a . It is not an assertion that the chosen alphabet size is minimal. Exact entropy cannot be smaller than outer entropy by definition, so the ordering in the corollary is the relevant restriction.

7. A LYAPUNOV CONDITION AND ENTROPY SEPARATION

7.1. A system-level obstruction. The next result uses a one-step inequality for a scalar function. It does not assume shrinking trajectory diameters or any positive invariance entropy. The cone formula supplies positive control entropy separately, under explicit algebraic and viability assumptions.

Theorem 7.1 (Common Lyapunov obstruction). *Let X, U, f, Q be as in section 2, and suppose $p \in Q$ satisfies $f(p, v) = p$ for every $v \in U$. Suppose there is a continuous function $V : Q \rightarrow [0, L]$, with $L = \max_Q V$, such that $V^{-1}(0) = \{p\}$, and a continuous nondecreasing function $\psi : [0, L] \rightarrow [0, L]$ with $\psi(0) = 0$ and $\psi(t) < t$ for $t > 0$, satisfying*

$$(7.1) \quad V(f(x, v)) \leq \psi(V(x)) \quad \text{whenever } x \in Q, v \in U, f(x, v) \in Q.$$

Then:

- (i) *all infinite feasible trajectories converge to p , uniformly over their initial states and their controls; every two such trajectories are asymptotic;*
- (ii) *the state-trajectory shift has $h_{\text{top}}(S) = 0$, and the uniform fiber entropy in (2.6) satisfies $h_{\text{fib}}(F \mid \pi) = 0$;*
- (iii) *$\pi(\Lambda_Q) = \mathcal{U}$ and*

$$h_{\text{top}}(F) = h_{\text{top}}(\sigma : \mathcal{U} \rightarrow \mathcal{U}) = \begin{cases} \log m, & |U| = m < \infty, \\ +\infty, & U \text{ infinite;} \end{cases}$$

(iv) every F -invariant Borel probability measure is supported on $\mathcal{U} \times \{p\}$. On that set π is a conjugacy to the input shift. In particular

$$h_\mu(F \mid \pi^{-1}\mathcal{B}_\mathcal{U}) = 0;$$

(v) the eventual core is

$$\bigcap_{n \geq 0} F^n(\Lambda_Q) = \mathcal{U} \times \{p\}.$$

If $Q \neq \{p\}$, then F is not surjective.

Every continuous self-map $G : Q \rightarrow Q$ obtained from a feasible memoryless feedback has zero topological entropy. All feasible trajectories of any fixed causal controller, whether continuous or not and whether or not it uses memory, satisfy the asymptotic conclusion in (i).

Proof. If $L = 0$, then $Q = \{p\}$, and all assertions follow directly; assume $L > 0$. The numbers $\psi^n(L)$ decrease to a fixed point of ψ . Continuity and strict decrease away from zero force this limit to be zero. Iterating (7.1) along a feasible trajectory gives

$$V(\varphi(n, x, u)) \leq \psi^n(L).$$

For any $\varepsilon > 0$, the compact set $\{x \in Q : d(x, p) \geq \varepsilon\}$, if nonempty, has a strictly positive minimum of V . Thus there are numbers $\beta_n \downarrow 0$ with

$$(7.2) \quad \sup_{(u,x) \in \Lambda_Q} d(\varphi(n, x, u), p) \leq \beta_n.$$

For example take the maximum distance to p over $V^{-1}([0, \psi^n(L)])$. This proves (i), including the quantification over different controls.

We give uniform covering arguments for (ii) and (iii) to avoid identifying different relative entropy conventions. Fix $0 < \varepsilon < 1$ and choose N such that $2\beta_k < \varepsilon/2$ for $k \geq N$. The finitely many maps $(x, u_0, \dots, u_{N-1}) \mapsto \varphi(k, x, u)$, $0 \leq k < N$, are uniformly continuous on the compact domain $Q \times U^N$. This remains true even though intermediate states for some controls may leave Q ; each finite-time image of the compact domain is compact, and f is continuous on $X \times U$. There exists $\delta > 0$, independent of the control prefix, such that

$$(7.3) \quad d(x, y) < \delta \implies \max_{0 \leq k < N} d(\varphi(k, x, u), \varphi(k, y, u)) < \varepsilon/2$$

for every common prefix.

Cover Q by finitely many sets of diameter less than δ , say C sets. Within one input fiber, two feasible initial states in the same such set have state distance less than $\varepsilon/2$ up to time $N - 1$ by (7.3). At later times their distance is less than $\varepsilon/2$ by (7.2). Their input coordinates are identical at all shifted times. Consequently every (n, ε) -separated subset of every fiber has cardinality at most C , uniformly in n and u . This proves $h_{\text{fib}}(F \mid \pi) = 0$.

For the state-trajectory system, (7.2) implies

$$\sup_{\xi \in \mathcal{T}_Q} d_{\mathcal{T}_Q}(S^n \xi, (p, p, \dots)) \leq \beta_n \longrightarrow 0.$$

Any continuous map on a compact metric space whose iterates converge uniformly to a constant has zero entropy: for a given separation scale, all sufficiently late images have diameter smaller than that scale, while uniform continuity of the finitely many earlier iterates gives a finite cover of bounded Bowen diameter for every horizon. Applying this argument proves $h_{\text{top}}(S) = 0$.

The set $\mathcal{U} \times \{p\}$ belongs to Λ_Q , and F restricted to it is conjugate to the full input shift. This proves surjectivity of π and the lower entropy bound in (iii). If $|U| = m < \infty$, choose ℓ so that $2^{-\ell} < \varepsilon/2$. For large n , group lift points by their first $n + \ell$ input symbols and by a member of the above finite state cover. Points in one group have close initial state evolution until time N , since their input prefixes agree and $n + \ell \geq N$. Their later feasible

state evolutions are close by (7.2). At each shifted time $0 \leq k < n$, agreement of the input prefix gives input distance at most $2^{-\ell} < \varepsilon/2$. Thus each group has Bowen diameter less than ε , and

$$s_n(F, \varepsilon) \leq C m^{n+\ell}.$$

This proves the upper bound $\log m$. If U is infinite, it contains finite subsets of every cardinality m . Each corresponding full shift at the apex is a compact invariant subsystem of entropy $\log m$. Letting $m \rightarrow \infty$ proves that both full input entropy and total lift entropy are infinite.

For (iv), regard V also as a function on the lift. Invariance of μ gives

$$0 = \int_{\Lambda_Q} (V - V \circ F) d\mu \geq \int_{\Lambda_Q} (V - \psi \circ V) d\mu \geq 0.$$

The last integrand is nonnegative and vanishes only when the state is p . Hence $\mu(\mathcal{U} \times \{p\}) = 1$. Projection and $u \mapsto (u, p)$ are inverse measurable maps on this full-measure set and intertwine the transformations. The input factor therefore contains the whole state information modulo μ , and its conditional Kolmogorov–Sinai entropy is zero. Conversely, any invariant input measure lifts in this way.

For (v), every point of $F^n(\Lambda_Q)$ has V -value at most $\psi^n(L)$, so a point in the intersection has state p . Conversely every (u, p) admits arbitrarily long prehistories in the lift: prepend any n input values to u and keep the state at p . This proves the core formula. Finally some $x \in Q$ attains $V(x) = L > 0$; it has an infinite feasible control by controlled invariance. Its lift point cannot be in $F(\Lambda_Q)$, whose V -values are at most $\psi(L) < L$. Hence F is not onto.

A feasible controller generates only trajectories already covered by (i), regardless of how it makes its choices. If it induces a continuous self-map G of Q , then $G^n \rightarrow p$ uniformly; the same bounded-Bowen-cover argument used for S gives $h_{\text{top}}(G) = 0$. $\square$

For cone systems, take $V(s, z) = s$, $L = 1$, and $\psi = R|_{[0,1]}$. On Q_B , zero radial coordinate occurs only at the apex and (7.1) holds with equality. The preceding theorem therefore applies to every system in theorem 3.1, independently of its spectral assumptions for positive entropy.

7.2. The entire digit lift.

Proposition 7.2 (Explicit coding of the true lift). *For the digit system of theorem 6.1, let $\Sigma = U^{\mathbb{N}_0}$ and define*

$$Y_i(u) = - \sum_{j=0}^{\infty} \frac{u_{j,i}}{q_i^{j+1}}, \quad Y(u) = (Y_1(u), \dots, Y_d(u)).$$

Then

$$(7.4) \quad H : \Sigma \times [0, 1] \longrightarrow \Lambda_Q, \quad H(u, s) = (u, (s, sY(u)))$$

is a homeomorphism, and

$$F \circ H = H \circ (\sigma \times R).$$

In particular the total lift entropy is $\log M$, its uniform input-fiber entropy is zero, its state-trajectory entropy is zero, and every invariant lift measure is $H_*(\nu \times \delta_0)$ for a shift-invariant probability measure ν on Σ .

Proof. Since $|u_{j,i}| \leq q_i - 1$, the series converge uniformly and $|Y_i(u)| \leq 1$; Y is continuous in the product topology. The identity

$$q_i Y_i(u) + u_{0,i} = Y_i(\sigma u)$$

shows that $H(u, s)$ is feasible and proves equivariance.

Conversely, let $(u, (s, z)) \in \Lambda_Q$ with $s > 0$, and put $y = z/s$. The normalized trajectory remains in $[-1, 1]^d$. For each coordinate, dividing the n -step equation by q_i^n gives

$$y_i = - \sum_{j=0}^{n-1} \frac{u_{j,i}}{q_i^{j+1}} + q_i^{-n} y_{n,i}.$$

The final term tends to zero. Thus $y = Y(u)$. If $s = 0$, the state is the apex and is $H(u, 0)$. This proves surjectivity onto the *entire* feasible lift, not merely onto a selected invariant subset. The input and radial coordinates recover u, s uniquely, so H is injective. A continuous bijection from the compact domain to the Hausdorff lift is a homeomorphism.

The entropy and measure assertions follow from theorem 7.1. Alternatively each fiber is parameterized by one radial interval with an equicontinuous family of iterates, giving a direct uniform finite Bowen cover. Under H , the eventual core is precisely $\Sigma \times \{0\}$; no assertion of surjectivity on $\Sigma \times [0, 1]$ is made. $\square$

The full shift on $M \geq 2$ symbols is a continuous surjection of a compact metric space with positive topological entropy. The classical theorem [1, Corollary 2.4(2)] applies on the apex subsystem and supplies Li–Yorke chaos there. Every one of its state projections is the same fixed state p . In a common input fiber, even the full lift trajectories are asymptotic, because their input coordinates agree at all times and their state coordinates converge to p . Thus total lift chaos and absence of state Li–Yorke pairs coexist without a contradiction.

Quantity for the finite digit family	Value
Exact invariance entropy on K_d or Q	$\sum_i \log q_i$
Outer and tracking invariance entropies	$\sum_i \max\{0, \log(\rho q_i)\}$
Total entropy of the true lift	$\log M$
Uniform entropy over the full input fibers	0
Entropy of the state-trajectory shift	0
Conditional KS entropy over input, any invariant measure	0
Entropy of any continuous feasible closed-loop state map	0

7.3. Feedback regularity and the control-set boundary.

Proposition 7.3 (A feedback regularity obstruction). *For the cone system in theorem 3.1, if $\chi^+(A) > 0$, no memoryless feedback $\kappa : Q_B \rightarrow U$ continuous at p can satisfy $f(x, \kappa(x)) \in Q_B$ for every $x \in Q_B$. If $U \supset -AB$, there is nevertheless a Borel feedback whose induced state map is the restriction of a global smooth map,*

$$G(s, z) = (R(s), 0).$$

Proof. Suppose such a continuous-at- p feedback exists and write $v_0 = \kappa(p)$. For every $y \in B$ and every $0 < s \leq 1$, feasibility implies $Ay + \kappa(s, sy) \in B$. Letting $s \downarrow 0$ and using closedness of B gives $Ay + v_0 \in B$ for all $y \in B$. Thus the affine map $T_0(y) = Ay + v_0$ sends B into itself. Iteration and subtraction give

$$A^n(B - B) = T_0^n(B) - T_0^n(B) \subset B - B.$$

Since $B - B$ is bounded and contains a neighborhood of zero, the operators A^n are uniformly bounded. This excludes eigenvalues of modulus greater than one and contradicts $\chi^+(A) > 0$.

If $U \supset -AB$, define $\kappa(s, sy) = -Ay$ for $s > 0$ and $\kappa(p) = 0$. This is Borel and has values in U . Substitution gives $G(s, z) = (R(s), 0)$ on Q_B , which extends smoothly to the ambient space. Its feedback values depend on direction at the apex; continuity of the induced state map must not be confused with continuity of the feedback itself. $\square$

For the nonlinear systems of theorem 3.2, the same obstruction to continuity of a viable feedback holds. Indeed feasibility at (s, sy) would give

$$qy + \kappa(s, sy) + P_{\kappa(s, sy)}(s, sy)/s \in [-1, 1].$$

The last term tends to zero uniformly in $y \in [-1, 1]$ and the finite input alphabet. Continuity at the apex would therefore force $q[-1, 1] + \kappa(p) \subset [-1, 1]$, impossible for $q > 1$.

The common Lyapunov theorem excludes state Li–Yorke chaos for *every* feasible controller in this family, not merely for the displayed resetting feedback. The scalar baseline in appendix B illustrates why a property of one chosen feedback is logically different from a conclusion quantified over all feasible inputs.

A control set ordinarily includes a controlled invariance condition together with approximate controllability: for every x in the set, the set is contained in the closure of the forward reachable set from x , with a maximality condition. The cones here fail approximate controllability. The radial coordinate of every forward trajectory starting at $s \in (0, 1]$ is $R^n(s) \leq s$, regardless of control and even if the other coordinates leave the cone. Points with larger radial coordinate cannot be approximated from such a state. Therefore Q_B is not a genuine control set. This is a substantive boundary, not a removable technical omission.

The equi-invariance/instability alternative discussed in the author version of [14] does not repair the missing implication: its instability condition allows a nearby state under a fixed selected input to leave a neighborhood of the constraint. The two trajectories need not both remain feasible. In the cone family, strong expansion of an error in $y = z/s$ is consistent with the fact that all *feasible* ambient states eventually become close. Neither a single escape nor ordinary sensitivity supplies recurrent proximality.

Likewise, return to Q_B alone is not an additional recurrence hypothesis here, since feasible trajectories remain there at every time. A condition requiring repeated return to a nondegenerate interior region bounded away from the apex would exclude the present construction. Recurrence entropy [24] studies a different catalogue problem involving bounded-time returns; it cannot by itself be read as a statement about two feasible state trajectories.

These distinctions do not assert a positive common-input chaos theorem for control sets. Approximate controllability, a selected input shift, or one departure estimate alone does not establish recurrent proximality and persistent separation of feasible state trajectories. The scalar calculation in appendix B gives a separate obstruction to identifying positive invariance entropy with common-input proximality.

8. FROM A FIXED TOLERANCE TO SHRINKING PRECISION

The common radial family and the globally contracting planar family use the same assignment rule, but impose different conditions on unsuccessful trajectories. The following elementary consequence isolates that distinction.

Proposition 8.1 (Fixed-tolerance costs under all-input contraction). *Let $Q \subset \mathbb{R}^d$ be compact and controlled invariant, let $0 \in Q$, and suppose every continuous controlled map fixes zero. Assume there are a norm V and $0 < \theta < 1$ such that*

$$V(f(x, v)) \leq \theta V(x) \quad (x \in \mathbb{R}^d, v \in U).$$

For each fixed $\varepsilon > 0$, there is a finite tracking catalogue for all horizons and all initial states in Q . Consequently

$$h_{\text{out}}(K, Q) = h_{\text{inv}}^+(K, Q) = 0 \quad (\emptyset \neq K \subset Q \text{ compact}).$$

All feasible state-trajectory pairs are uniformly asymptotic, and the zero state and fibre conclusions of Theorem 7.1 apply.

Proof. By equivalence of norms and compactness, choose J such that any two trajectories starting in Q , under any inputs, are at Euclidean distance less than $\varepsilon/2$ at every time $k \geq J$. For each $x \in Q$, select one exactly feasible infinite input u^x . Continuity at the finitely many times $0, \dots, J$ gives a relative neighbourhood O_x in Q on which the trajectory under u^x remains within ε of the reference from x during those times. Take a finite subcover $O_{x_1}, \dots, O_{x_M}$. The pairs (x_j, u^{x_j}) work at the early times by construction and at all later times by the global norm estimate, whether the actual trajectory remains feasible or not. The number M is independent of the horizon. The outer inequality (2.4) gives the second zero rate. The Lyapunov choice $V|_Q$ and $\psi(t) = \theta t$ gives the final assertion. $\square$

The nonlinear equal-entropy cone examples have converging *feasible* trajectories but need not satisfy the global estimate in this proposition. There is therefore no conflict between their positive fixed-tolerance rates and the zero rates of the globally contracting systems. For the latter, the next sections shrink the tolerance with the horizon and, subsequently, also shrink the admissible failure probability. None of these changes identifies a state entropy with a control cost.

9. SHRINKING PRECISION AND RELIABILITY: ASSUMPTIONS AND RESULTS

9.1. Catalogue and precision. Let

$$Q = \{(s, z) \in \mathbb{R}^2 : 0 \leq s \leq 1, |z| \leq s\}, \quad \text{area}(Q) = 1.$$

For an infinite binary input $u = (u_1, u_2, \dots)$ set $F_u^0 = \text{id}$ and $F_u^k = F_{u_k} \circ \dots \circ F_{u_1}$. The one-based input indexing here shifts the zero-based convention in Section 2 by one and changes no count. All binary inputs are allowed. Distances are Euclidean, and all logarithms are natural. For a nonempty compact $K \subset Q$ define

$$(9.1) \quad A_u(n, \delta; K) = \{x \in K : \text{dist}(F_u^k x, Q) < \delta \text{ for every } 0 \leq k \leq n\}.$$

Let $r_F(n, \delta; K)$ be the smallest number of infinite inputs whose sets (9.1) cover K . The assignment may depend on the actual initial state. Exit and reentry are allowed subject to every displayed constraint. Only the first n letters affect this count. Controlled feasibility makes it finite. In the notation of Section 2, this is exactly $r_{\text{inv}}(n, K, Q^\delta)$ for these planar maps; the symbol r_F records the actual controlled system.

A precision sequence means $\delta_n > 0$, $\delta_n \rightarrow 0$ and $-\log \delta_n/n \rightarrow \gamma$ for a finite $\gamma \geq 0$. It need not be monotone.

Let $\mu = \text{area}|_Q$, which is a probability since $\text{area}(Q) = 1$. For $0 < e < 1$, define

$$(9.2) \quad r_e(n, \delta; Q) = \min \left\{ \#\mathcal{U} : \mu \left(\bigcup_{u \in \mathcal{U}} A_u(n, \delta; Q) \right) \geq 1 - e \right\}.$$

The minimum is over finite catalogues of unrestricted binary inputs. It exists because the full-cover count is finite. A failure sequence has $0 < e_n < 1$, $e_n \rightarrow 0$ and $-\log e_n/n \rightarrow \alpha > 0$. Both sequences may be nonmonotone. The successful physical subset in (9.2) may change with n ; it is not a fixed compact initial set.

9.2. First-order data and global assumptions. Fix

$$(9.3) \quad a_0, a_1 \in (0, 1), \quad a_0 + a_1 = 1, \quad a_0 \neq a_1, \quad \beta_0, \beta_1 > 1,$$

and set $r_i = a_i^{\beta_i}$, $t_i = r_i/a_i < 1$, $a_* = \min_i a_i$, $a^* = \max_i a_i$ and $h = -\sum_i a_i \log a_i$. Write

$$(9.4) \quad A_0 = \begin{pmatrix} r_0 & 0 \\ t_0 a_1 & t_0 \end{pmatrix}, \quad A_1 = \begin{pmatrix} r_1 & 0 \\ -t_1 a_0 & t_1 \end{pmatrix}.$$

For the planar results in the rest of the main text, the standing assumptions on the actual maps are:

- (i) $F_0, F_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ are C^2 maps, $F_i(0) = 0$ and $DF_i(0) = A_i$.

(ii) $Q \subset F_0^{-1}Q \cup F_1^{-1}Q$.

(iii) For some $\Lambda \geq 1$ and $0 < \theta < 1$, the norm $V(s, z) = \max\{\Lambda|s|, |z|\}$ satisfies

$$(9.5) \quad V(F_i x) \leq \theta V(x) \quad (x \in \mathbb{R}^2, i = 0, 1).$$

(iv) Each critical set

$$(9.6) \quad \mathcal{C}_i = \{x \in Q : \det DF_i(x) = 0\}$$

has area zero.

The second assumption is one-step controlled invariance; unrestricted concatenation yields an exactly feasible infinite input from each point of Q . The third concerns contraction toward the origin, not distances between two arbitrary trajectories. In particular it controls appended tails even outside Q . The fourth supplies noncollapse along feasible finite transients; it allows folds and requires no condition on critical points outside Q . Radial motion may depend on angle. No accumulated distortion, word-area bound or catalogue comparison is assumed.

Both matched and mismatched subfamilies are nonempty. For the linear maps A_i , the angular domains $[-1, a_0 - a_1]$ and $[a_0 - a_1, 1]$ cover every fibre. Put $t = \max_i t_i < 1$ and choose $\Lambda = 1 + 4t/(1 - t)$. Then $V(A_i x) \leq \max\{\max_i r_i, t(1 + 1/\Lambda)\}V(x)$ globally, with factor less than one, and $\det A_i = r_i^2/a_i > 0$.

For words write

$$\lambda_w = \prod_{j=1}^{|w|} a_{w_j}, \quad P_w = \prod_{j=1}^{|w|} r_{w_j},$$

with empty products one. Define

$$(9.7) \quad \chi_R = - \sum_i a_i \log r_i > h, \quad \gamma_0 = - \log \max_i r_i > 0, \quad r_0^D + r_1^D = 1.$$

The root D is unique in $(0, 1)$, since the left side decreases from two to zero and $r_0 + r_1 < 1$. Also $\chi_R \geq \gamma_0$.

9.3. Probability costs. Write H_b for binary entropy. In the matched case $\beta_0 = \beta_1 = \beta$, the angular cost and spectrum are

$$(9.8) \quad H_b(p) = -p \log p - (1 - p) \log(1 - p), \quad \chi(p) = -p \log a_0 - (1 - p) \log a_1,$$

$$(9.9) \quad S(\gamma) = \max_{p \in [0, 1]} H_b(p) \min\{1, \gamma/[\beta \chi(p)]\},$$

where $0 \log 0 = 0$. In particular $H_b(a_0) = \chi(a_0) = h$.

$$(9.10) \quad I_a(p) = \chi(p) - H_b(p), \quad L_a(\alpha) = \max_{I_a(p) \leq \alpha} H_b(p).$$

The function I_a is relative entropy with respect to (a_0, a_1) ; L_a is the classical binary source-coding reliability function.

Theorem 9.1 (Precision–reliability transfer). *Assume the standing hypotheses with $\beta_0 = \beta_1 = \beta$, and strengthen the critical-set condition to*

$$(9.11) \quad \det DF_i(x) \neq 0 \quad (x \in Q, i = 0, 1).$$

For every fixed such system, every finite $\gamma \geq 0$, every $\alpha > 0$, and every precision and failure sequence of those exponents,

$$(9.12) \quad \lim_{n \rightarrow \infty} \frac{\log r_{e_n}(n, \delta_n; Q)}{n} = \min\{S(\gamma), L_a(\alpha)\}.$$

No global injectivity or global inverse is required.

For a finite catalogue, enumerate its inputs and assign a successful state to its first successful input. The sets are Borel, so this is a measurable initial-state encoder. A fixed-length index sent once at time zero has minimum bit length $\lceil \log_2 r_e \rceil$; the asymptotic bit rate is (9.12) divided by $\log 2$. There are no later messages. This interpretation does not identify the catalogue rate with a causal feedback data-rate or a channel capacity. The bit conversion is a definition-level consequence.

Theorem 9.2 (A null-critical transient changes reliability). *For $a_0 = 2^{-10}$, $a_1 = 1 - 2^{-10}$ and $\beta = 4096$, there is a fixed injective C^2 pair satisfying the standing matched hypotheses, with critical sets of area zero, such that*

$$(9.13) \quad r_F(n, \delta; Q) = r_A(n, \delta; Q) \quad (n \geq 1, \delta > 0).$$

It maps no positive-area subset of Q into an area-zero set. Nevertheless, for $\gamma = -\beta \log(a_0 a_1)$ and $\alpha = 1/8$, every pair of sequences with these exponents satisfies

$$(9.14) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \frac{1}{2} \log 2 > \min\{S(\gamma), L_a(\alpha)\}.$$

Condition (9.11) is a sufficient condition for (9.12). The counterexample rules out replacing it merely by critical sets of area zero in a universal transfer theorem. It does not show that every system with critical points fails, or give a necessary and sufficient probability criterion.

9.4. Ordinary precision costs and fixed initial sets. The following results use only the standing null-critical hypothesis. They both describe the baseline precision cost and distinguish it from the exponentially reliable cost above.

Theorem 9.3 (Precision spectrum and initial-set dependence). *Suppose $\beta_0 = \beta_1 = \beta$ in the standing assumptions. For every fixed system, every finite $\gamma \geq 0$ and every precision sequence of exponent γ ,*

$$(9.15) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = S(\gamma).$$

Every fixed positive-area compact $K \subset Q$ satisfies

$$(9.16) \quad \min\{h, \gamma/\beta\} \leq \liminf_n \frac{\log r_F(n, \delta_n; K)}{n} \leq \limsup_n \frac{\log r_F(n, \delta_n; K)}{n} \leq S(\gamma).$$

Thus the rate exists and is γ/β for $0 \leq \gamma \leq \beta h$. For each $\zeta \in (0, 1)$ there is one fixed compact $K_\zeta \subset Q$ with $\text{area}(K_\zeta) > 1 - \zeta$ such that, simultaneously for every finite $\gamma \geq 0$ and every precision sequence of that exponent,

$$(9.17) \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \min\{h, \gamma/\beta\}.$$

The set is chosen before the exponent, sequence and horizon.

Corollary 9.4 (Closed spectrum and two thresholds). *Under the hypotheses of Theorem 9.3, put $a = a_* \in (0, 1/2)$, $c_b = -\log(1 - a)$ and $d_b = \log((1 - a)/a) > 0$. The common-cost boundary and saturation boundary are respectively*

$$(9.18) \quad \gamma_* = \beta h, \quad \gamma_{\text{sat}} = -\frac{\beta}{2} \log(a_0 a_1) = \beta(c_b + d_b/2) > \gamma_*.$$

The full-set spectrum is

$$(9.19) \quad S(\gamma) = \begin{cases} \gamma/\beta, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b), & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

The exact gap between Q and the single set K_ζ in the theorem is

$$(9.20) \quad \Delta(\gamma) = S(\gamma) - \min\{h, \gamma/\beta\} = \begin{cases} 0, & 0 \leq \gamma \leq \gamma_*, \\ H_b((\gamma/\beta - c_b)/d_b) - h, & \gamma_* < \gamma < \gamma_{\text{sat}}, \\ \log 2 - h, & \gamma \geq \gamma_{\text{sat}}. \end{cases}$$

It is strictly positive exactly when $\gamma > \gamma_*$. The full-set rate equals $\log 2$ exactly when $\gamma \geq \gamma_{\text{sat}}$.

The first threshold concerns equality of rates across all positive-area compact initial sets; the second concerns the full-set catalogue alone. The theorem does not classify high-side rates of arbitrary compact initial sets or assert that only two such rates are possible.

Theorem 9.5 (First-order matching criterion). *For every fixed system satisfying the standing assumptions, the following property holds if and only if $\beta_0 = \beta_1$: there are $\gamma_1 > 0$ and a function $c : (0, \gamma_1) \rightarrow [0, \infty)$ such that, for every fixed positive-area compact $K \subset Q$, every $\gamma \in (0, \gamma_1)$, and every precision sequence of exponent γ ,*

$$\lim_n n^{-1} \log r_F(n, \delta_n; K) = c(\gamma).$$

Theorem 9.6 (Low-precision mismatch). *Suppose $\beta_0 \neq \beta_1$. For each $\zeta \in (0, 1)$ there is one compact $K_\zeta \subset Q$, of area greater than $1 - \zeta$, such that simultaneously for every $0 < \gamma < \gamma_0$ and every precision sequence of that exponent,*

$$(9.21) \quad \lim_n \frac{\log r_F(n, \delta_n; Q)}{n} = \gamma D, \quad \lim_n \frac{\log r_F(n, \delta_n; K_\zeta)}{n} = \gamma h / \chi_R < \gamma D.$$

The set is selected before the exponent, sequence, and horizon. Every other fixed positive-area compact $K \subset Q$ has lower rate at least $\gamma h / \chi_R$ in this range.

9.5. The physical coverage estimate. The following estimate bounds the mass covered by one arbitrary input, without selecting an exact coding. It supplies the lower rate for positive-area initial sets in both cases. Its regular compact set also provides the single near-full set used in the upper bounds. These are fixed-loss estimates: their $C_T \delta$ term is not assigned to an exponentially small probability budget.

Theorem 9.7 (Area covered by an arbitrary approximate input). *Under the standing assumptions, whether matched or mismatched, there is a fixed integer $J \geq 1$ with the following property. For every positive-area compact $K \subset Q$ and every $0 < \varepsilon < \text{area}(K)$, one can choose a compact $T \subset K \setminus \{0\}$ with $\text{area}(T) > \text{area}(K) - \varepsilon$ and $C_T < \infty$ so that for every $e \in (0, h)$ there is $C_{T,e} < \infty$ such that*

$$(9.22) \quad \text{area}(T \cap A_u(n, \delta; K)) \leq C_T \delta + C_{T,e} \left[e^{-(h-e)m} + \delta e^{(\chi_R - h + 2e)m} \right]$$

for every input u , $n \geq J$, $0 \leq m \leq n - J$ and $0 < \delta \leq 1$. The integer depends only on the fixed system. The set and C_T precede e, n, δ, u ; the second constant may also depend on e . In particular, for $K = Q$ the set can have area greater than $1 - \zeta$ for any prescribed $\zeta \in (0, 1)$.

10. LOCAL FEASIBLE GEOMETRY

We first control complete feasible regions near the origin. The angular and radial products describe different scales, so they remain separate throughout this section. The first-exit estimate will then apply to any input, even one that is only approximately feasible.

Use (s, y) with $z = sy$, only for $s > 0$. Define

$$R_i(s, y) = F_i^s(s, sy)/s, \quad N_i(s, y) = F_i^z(s, sy)/s, \quad T_i = N_i/R_i.$$

The integral formula $F_i(s, sy)/s = \int_0^1 DF_i(ts, tsy)(1, y) dt$ extends R_i, N_i as C^1 functions to $s = 0$. At zero

$$(10.1) \quad R_i(0, y) = r_i, \quad L_0(y) := T_0(0, y) = (y + a_1)/a_0, \quad L_1(y) := T_1(0, y) = (y - a_0)/a_1.$$

We choose the local constants from the maps as follows. Let L bound the Euclidean bilinear operator norm of both second derivatives on $\overline{B}(0, 2)$, and let $r_* = \min r_i$, $r^* = \max r_i$, $a^* = \max a_i$, $a_* = \min a_i$. Take $B = 10^4(1 + L)/r_*^3$, and shrink $\rho > 0$ so that

$$(10.2) \quad \rho \leq \min \left\{ \frac{1}{4}, \frac{r_*}{8(1 + L)}, \frac{1 - r^*}{8B}, \frac{1/a^* - 1}{16B}, \frac{1}{16B}, \frac{a_*}{16B} \right\}.$$

Taylor's integral formula gives $|R_i - r_i| \leq 2Ls$, $|\partial_s R_i|, |\partial_s N_i| \leq 2L$, and $|\partial_y R_i| \leq 2Ls$, $|\partial_y N_i - t_i| \leq 2Ls$. The same bounds apply with the appropriate constant linear term in N_i . In particular $R_i \geq r_*/2$. Dividing these formulas by R_i , and differentiating the quotient, gives, with the generous common constant B ,

$$(10.3) \quad \begin{aligned} |R_i - r_i| &\leq Bs, & |\log(R_i/r_i)| &\leq Bs, \\ |\partial_s \log R_i| &\leq B, & |\partial_y \log R_i| &\leq Bs, \\ |T_i - L_i| &\leq Bs, & |\partial_s T_i| &\leq B, & |\partial_y T_i - 1/a_i| &\leq Bs. \end{aligned}$$

For example $|N_i| \leq 3$ on this rectangle; quotient derivatives use $R_i^{-2} \leq 4/r_*^2$. Thus the displayed choice of B dominates all the constants in (10.3). These are derived local inequalities, not extra hypotheses.

Set $\tilde{r} = r^* + B\rho < 1$ and $\hat{r} = r^* + 2B\rho < 1$. On an exactly feasible local orbit, $s_k \leq \tilde{r}^k s_0$. The bounds in (10.2) also give $T_1(s, -1) < -1$ and $T_0(s, 1) > 1$. Controlled invariance therefore forces

$$(10.4) \quad T_0(s, -1) \geq -1, \quad T_1(s, 1) \leq 1.$$

Equality is not required in (10.4). The increasing functions $T_i(s, \cdot)$ have unique cut points $\vartheta_0(s)$ with $T_0 = 1$ and $\vartheta_1(s)$ with $T_1 = -1$. They equal $a_0 - a_1 + O(s)$. Their feasible intervals are respectively $[-1, \vartheta_0(s)]$ and $[\vartheta_1(s), 1]$. Controlled invariance gives $\vartheta_1 \leq \vartheta_0$. Thus any local overlap has angular width $O(s)$ and physical width $O(s^2)$. Positivity of the overlap is not assumed.

10.1. Complete intervals with truncated images. Consider a radial graph $s = S(y)$, on any closed subinterval of $[-1, 1]$, with $|S'/S| \leq 1$. Put $k = S'/S$. For either input its terminal angular derivative along the graph is

$$J_i = \partial_y T_i + \partial_s T_i S' = 1/a_i + O(BS) > 0.$$

The new radial graph has logarithmic slope

$$(10.5) \quad k_{\text{new}} = \frac{k + (\partial_s \log R_i)S' + \partial_y \log R_i}{J_i}.$$

Its numerator is at most $1 + 2B\rho$ in absolute value and its denominator at least $1/a^* - 2B\rho$. By (10.2), $|k_{\text{new}}| \leq 1$. Piecewise C^1 graphs obey the same assertion almost everywhere.

For a fixed initial radius $0 < s_0 \leq \rho$, start with the horizontal graph $S \equiv s_0$. Let $I_w(s_0)$ be the set of initial angles whose trajectory under a finite word w stays in Q at every intermediate time. It is a closed interval, possibly empty or a point. Here is the induction that checks all constraints. The current angular image is an interval, with a graph obeying (10.5). Extend that graph constantly to the two sides of its domain. The extension is positive, bounded by ρ , and obeys the logarithmic-slope bound almost everywhere. Both $T_i(S(y), y)$ are increasing. By (10.4), control 0 cannot violate the lower angular boundary, and control 1 cannot violate the upper one. The other boundary imposes one cut. Pulling that cut back through the increasing preceding angular map leaves an interval. No terminal

map is asserted to be onto $[-1, 1]$. The two allowed children cover the parent, by controlled invariance, but either child can be empty.

Along any such feasible graph, (10.3) and the geometric decay of s_j give constants independent of w, s_0 and its initial angle:

$$(10.6) \quad C_R^{-1} s_0 P_w \leq s_{|w|} \leq C_R s_0 P_w, \quad C_A^{-1} \lambda_w^{-1} \leq \frac{dY_w}{dy_0} \leq C_A \lambda_w^{-1}.$$

For instance one may take $C_R = \exp(B\rho/(1 - \bar{r}))$, $C_A = \exp(4B\rho/(1 - \bar{r}))$: $|a_i J_i - 1| \leq 2Bs_j \leq 1/8$, so its logarithm has modulus at most $4Bs_j$. The first estimate sums $\log(R_i/r_i)$; the second sums the angular derivatives. They are independent product estimates. Only in the matched case is $P_w = \lambda_w^\beta$. Integration over the *actual* angular image implies

$$(10.7) \quad |I_w(s_0)| \leq 2C_A \lambda_w, \quad |\mathcal{I}_w| \leq C_A \rho^2 \lambda_w, \quad \mathcal{I}_w = \{(s, sy) : 0 < s \leq \rho, y \in I_w(s)\}.$$

The Jacobian of $(s, y) \mapsto (s, sy)$ is s . There is no positive lower bound on $|I_w|$; some fixed-radius slices in Section 17.1 are empty.

For a point interval the length bound is zero. Derivative comparisons on nondegenerate intervals extend to their endpoints by continuity; degenerate fibres remain in all covers and do not contribute area.

10.2. The first true exit. For a fixed input, partition its covered states by the first time at which its trajectory actually leaves Q . Up to that time the prefix is exactly feasible, so the preceding interval estimates apply. The permitted tolerance restricts the states that can survive this first exit.

Suppose the prefix w of length $j - 1$ is exactly feasible and the next prescribed input i is the first one to leave Q . On the current graph the map T_i is increasing with derivative at least $1/(2a_i)$. The radial output is at least $cs_0 P_w$, by (10.3), (10.6), with $c > 0$ fixed. If this wrong step nonetheless has distance less than δ from Q , then

$$(10.8) \quad (|z_j| - s_j)_+ < 2\delta.$$

This follows from the Euclidean Lipschitz linear functionals $z - s$ and $-z - s$. The output radius is positive and at most ρ , so the violated constraint is angular, not the upper radius. Consequently the wrong part of the terminal angular interval has diameter at most $C\delta/(s_0 P_w)$. By (10.6), its initial-angle preimage has length at most

$$C\delta \lambda_w / (s_0 P_w).$$

After integration against $s_0 ds_0$, its physical area is at most

$$(10.9) \quad C\delta \lambda_w / P_w.$$

This argument also covers a point interval or a terminal interval lying wholly beyond the cut: monotonicity bounds the diameter of the part satisfying (10.8), without requiring a cut point inside the actual image. It does not require the wrong trajectory to remain outside afterwards. Any later reentry cannot erase constraint (10.8) at its first exit. A different exact feasible coding is not counted as an exit and has no role in (10.9).

11. FINITE TRANSIENTS AND ACTUAL COVERED AREA

Common contraction gives a uniform time of entry into the local region. To transport its area estimates back to the initial set, we remove a small area once and retain all feasible transient choices on the remaining compact window. This is where the null-critical-set hypothesis is used; a global inverse is not required.

For a finite word w , let

$$(11.1) \quad D_w = \{x \in Q : F_{w|k}x \in Q \ (1 \leq k \leq |w|)\}, \quad D_\emptyset = Q.$$

These sets are compact. Write w^- for the word with its last letter removed. At a first true exit under w , all derivative inputs still lie in Q , so the relevant domain is D_{w^-} . The last output need not belong to Q .

Lemma 11.1 (Null pullbacks along feasible prefixes). *For every Lebesgue null set $B \subset \mathbb{R}^2$, every control i , and every finite word w ,*

$$(11.2) \quad \text{area}(Q \cap F_i^{-1}B) = 0, \quad \text{area}(D_w \cap F_w^{-1}B) = 0.$$

For each nonempty word the compact set

$$(11.3) \quad Z_w = D_{w^-} \cap \{x : \det DF_w(x) = 0\}$$

also has area zero.

Proof. The inverse function theorem supplies an open inverse chart at every point of $Q \setminus \mathcal{C}_i$. A countable family of these charts covers that set. Exhaust each image by relatively compact subcharts; its C^1 inverse is Lipschitz on sufficiently small compact neighbourhoods. A Lipschitz map preserves null sets: cover a bounded null set by squares with arbitrarily small total area, and cover their images by discs whose total area is bounded by a constant times that sum. Thus every chart pullback of B is null. The omitted $\mathcal{C}_i$ is null by (9.6). A Borel null superset handles an arbitrary Lebesgue null set.

For $w = vi$, exact feasibility gives

$$D_w \cap F_w^{-1}B \subset D_v \cap F_v^{-1}(Q \cap F_i^{-1}B).$$

Induction proves the second assertion of (11.2), including the empty word. Finally, on D_{w^-} the chain rule expresses $\det DF_w$ as a product of determinants evaluated at states in Q . If the factor at time j vanishes, the initial state belongs to

$$D_{w|j-1} \cap F_{w|j-1}^{-1}\mathcal{C}_{w_j}.$$

This is null by (11.2). Their finite union contains Z_w , which is compact by continuity and closedness of the feasibility domain. $\square$

Choose $J \geq 1$, depending only on the fixed system, so that

$$(11.4) \quad \theta^J \sup_Q V \leq \rho.$$

Every exactly feasible J -prefix ends in $Q_\rho = Q \cap \{s \leq \rho\}$.

Lemma 11.2 (One regular compact window). *For every positive-area compact $K \subset Q$ and $0 < \varepsilon < \text{area}(K)$, there is a compact $T \subset K \setminus \{0\}$, with $\text{area}(T) > \text{area}(K) - \varepsilon$, and $L_T < \infty$ such that*

$$(11.5) \quad \text{area}(T \cap D_{w^-} \cap F_w^{-1}B) \leq L_T \text{area}(B)$$

for all $1 \leq |w| \leq J$ and all Lebesgue measurable $B \subset \mathbb{R}^2$. On the same set, for every $e \in (0, h)$ there is $D_e \geq 1$ such that every local exactly feasible tail v , after any exactly feasible J -prefix from T , satisfies

$$(11.6) \quad \begin{aligned} D_e^{-1}e^{-(h+e)|v|} &\leq \lambda_v \leq D_e e^{-(h-e)|v|}, \\ D_e^{-1}e^{-(\chi_R+e)|v|} &\leq P_v \leq D_e e^{-(\chi_R-e)|v|}. \end{aligned}$$

The set and L_T precede e , every input, tolerance and horizon.

Proof. For $d > 0$, the total weight $\sum \lambda_v$ of length- m words with $|\#0/m - a_0| > d$ is at most $2e^{-2d^2m}$. Indeed, for a Bernoulli variable X of mean a_0 , the second derivative of $\log \mathbb{E}e^{t(X-a_0)}$ is a tilted variance, at most $1/4$. Its value and derivative at zero vanish, so it is at most $t^2/8$. Exponential Markov with $t = 4d$ and $t = -4d$ gives the bound. By (10.7), the unions of the corresponding complete local feasible regions have summable area. Borel–Cantelli for rational $d > 0$ gives a Borel null set $B_{\text{loc}} \subset Q_\rho$, including the origin,

outside which all feasible length- m prefixes have zero frequency tending uniformly to a_0 . Overlaps do not invalidate this union bound. Controlled invariance supplies at least one feasible prefix at every length.

Lemma 11.1 shows that

$$B_J = \bigcup_{|\sigma|=J} (D_\sigma \cap F_\sigma^{-1} B_{\text{loc}})$$

is null. The finite union $Z_J = \bigcup_{1 \leq |w| \leq J} Z_w$ is a compact null set. On $Q \setminus B_J$, the maximum frequency deviation over all feasible transient choices and all their feasible length- m tails tends to zero. It is measurable: for each m there are finitely many words and their complete feasibility domains are closed. Egorov on K , followed by removal of $B_J \cup Z_J \cup \{0\}$ and compact inner approximation, gives T of the stated area with uniform convergence. Allocate less than $\varepsilon/2$ loss to Egorov and less than $\varepsilon/2$ to inner approximation. Both logarithmic products are affine functions of frequency. Uniform convergence gives (11.6); finitely many early lengths are absorbed in D_e . This selects one set before all precision choices, rather than a set for each exponent.

For a word w with $1 \leq |w| \leq J$, the compact set $M_w = T \cap D_{w-}$ is disjoint from Z_w . The composition F_w is therefore regular at every point of M_w . Cover it by finitely many open inverse charts $U_{w,k}$, with compact closures inside regular inverse charts. Change of variables on each chart gives

$$\text{area}(M_w \cap U_{w,k} \cap F_w^{-1} B) \leq \left(\sup_{\bar{U}_{w,k}} |\det DF_w|^{-1} \right) \text{area}(B).$$

The supremum is finite. Sum over the charts and take the maximum over the finite word family, omitting empty M_w 's, to obtain (11.5). Chart overlaps only enlarge the bound; every local inverse branch meeting M_w is included. $\square$

The lemma provides an inverse-area bound on a fixed regular window, not a Jacobian lower bound on all Q or a global multiplicity bound. Its constants may depend on the prescribed area loss. The upper catalogues below still cover every actual state, including the critical states omitted from this window.

Proof of Theorem 9.7. Choose J, T, L_T from Lemma 11.2. Let $Q_\delta = \{x : \text{dist}(x, Q) < \delta\}$. For $0 < \delta \leq 1$, the tubes around the three edges of Q give

$$\text{area}(Q_\delta \setminus Q) \leq C_Q \delta, \quad C_Q = 2(2 + 2\sqrt{2}) + 3\pi.$$

If a state in $T \cap A_u(n, \delta; K)$ first truly exits at $j \leq J$, put $w = u|j$. Its earlier states lie exactly in Q , and it belongs to

$$T \cap D_{w-} \cap F_w^{-1}(Q_\delta \setminus Q).$$

By (11.5), its class has area at most $L_T C_Q \delta$. Summing over the J possible first-exit times gives

$$(11.7) \quad \text{area}\{x \in T \cap A_u(n, \delta; K) : \text{first true exit occurs by } J\} \leq C_T \delta, \quad C_T = J L_T C_Q.$$

All derivative inputs here are in Q ; no nonsingularity condition is needed at the last output, which may be outside.

For the remaining states, $\sigma = u|J$ is exactly feasible. Transport them by F_σ to Q_ρ , and use (11.5) to pull back endpoint area estimates. If the next m prescribed digits are exactly feasible for at least one transported state, (10.7) and (11.6) bound the area of the survival class by $C_{T,e} e^{-(h-e)m}$. If no such state exists, that class is empty. For a first true local exit at time $j \leq m$, the prescribed $(j-1)$ -prefix is feasible for a transported state whenever the class is nonempty. Its pulled-back area, by (10.9), (11.5) and both products in (11.6), is at most

$$C_{T,e} \delta e^{(\chi_R - h + 2e)(j-1)}.$$

Since $\chi_R > h$, the sum is bounded by a fixed multiple of $\delta e^{(\chi_R - h + 2e)m}$. Together with (11.7) and exact survival, this proves (9.22). A switch between feasible codes was never counted as an exit. Every covered state satisfies the constraint at its first true exit, regardless of subsequent reentry. $\square$

12. PHYSICAL TRAJECTORIES WITH A WRONG-CONTROL MARGIN

The area estimate concerns positive-area mass. For the full constraint we also need actual states whose controls cannot be replaced by approximate alternatives. We construct them at one fixed positive radius for each run bound, then force prefixes while their physical margin remains larger than the tolerance.

Lemma 12.1 (Bounded-run realisation). *For each integer $L \geq 2$, there is a positive radius s_L and $d'_L > 0$ such that, for every $0 < s \leq s_L$, every infinite binary sequence whose successive runs have length at most L is realised by an actual exactly feasible orbit starting at (s, sy_0) . At every step of that orbit the alternative control is genuinely infeasible, with an angular excess at least d'_L . The initial-angle centre can be chosen Lipschitz in s , with a constant uniform over the sequences. The radius bound and margin are uniform over the sequences. No such assertion is made for unbounded runs or all constant words.*

Proof. The inverse reference maps are $f_0(y) = a_0y - a_1$, $f_1(y) = a_1y + a_0$. For any fixed infinite input v , successive inverse compositions give a unique reference bounded orbit $y_k^0 \in [-1, 1]$. If runs have length at most L , the next opposite digit is seen within $L + 1$ digits. Since $f_0(y) + 1 = a_0(y + 1)$, $1 - f_1(y) = a_1(1 - y)$, and each opposite digit inserts an endpoint distance at least $2a_*$,

$$(12.1) \quad |y_k^0| \leq 1 - d_L, \quad d_L = 2a_*^{L+1} > 0.$$

Let $b = d_L/4$ and consider the complete closed ball of sequences $\|y - y^0\|_\infty \leq b$ in ℓ^∞ . For each such sequence and a positive fixed initial radius s_* , recursively define $s_{k+1} = s_k R_{v_{k+1}}(s_k, y_k)$. These radii stay below $\bar{r}^k s_*$, independently of whether the angular recursion has yet been imposed. The scalar map $s \mapsto s R_i(s, y)$ has derivative bounded in absolute value by $\hat{r} < 1$, and its y derivative is bounded by Bs^2 , since $R_i \leq \bar{r} < 1$ and (10.3) bounds its logarithmic angular derivative. Thus two candidate sequences give

$$(12.2) \quad \sup_k |s_k(y) - s_k(\tilde{y})| \leq C_s s_*^2 \|y - \tilde{y}\|_\infty, \quad C_s = B/(1 - \hat{r}).$$

Write $e_i(s, y) = T_i(s, y) - L_i(y)$. By (10.3), $|e_i| \leq Bs$, $|\partial_s e_i| \leq B$, and $|\partial_y e_i| \leq Bs$. Define

$$(12.3) \quad (\mathcal{B}y)_k = f_{v_{k+1}}(y_{k+1}) - a_{v_{k+1}} e_{v_{k+1}}(s_k(y), y_k).$$

Choose $s_* \leq \rho$ so small that

$$(12.4) \quad a^* B s_* \leq (1 - a^*)b/2, \quad a^* B(s_* + C_s s_*^2) < (1 - a^*)/2.$$

Then (12.3) maps the ball into itself and has Lipschitz constant at most $a^* + a^* B(s_* + C_s s_*^2) < 1$. The contraction mapping theorem gives a fixed sequence. Its equation is exactly $y_{k+1} = T_{v_{k+1}}(s_k, y_k)$, not a formal symbolic substitution. It has $|y_k| \leq 1 - 3d_L/4$, so the constructed physical trajectory is exactly feasible for all times.

To check the alternative control, suppose the chosen digit is zero. Then $T_0(s_k, y_k) = y_{k+1} \leq 1 - 3d_L/4$, and (10.3) gives

$$y_k \leq a_0 - a_1 - 3a_0 d_L/4 + a_0 B s_k.$$

Substitution into $T_1 = L_1 + e_1$ yields

$$T_1(s_k, y_k) \leq -1 - \frac{3a_0 d_L}{4a_1} + B(1 + a_0/a_1)s_k.$$

For chosen digit one, the symmetric computation gives

$$T_0(s_k, y_k) \geq 1 + \frac{3a_1 d_L}{4a_0} - B(1 + a_1/a_0)s_k.$$

Shrink s_* once more, before considering any particular sequence, so that these last error terms are at most $a_* d_L/(4a_*)$. Both wrong angular excesses are then at least $d'_L = a_* d_L/(2a_*)$. Set $s_L = s_*$. All choices depend only on L and the fixed system. The same estimates hold for every initial radius $0 < s \leq s_L$. For a fixed candidate sequence, scalar contraction gives $\sup_k |s_k(s, y) - s_k(s', y)| \leq |s - s'|$. If $\kappa_L < 1$ is the common contraction constant of (12.3), the fixed-point equations therefore give

$$(12.5) \quad \|y^v(s) - y^v(s')\|_\infty \leq \frac{a^* B}{1 - \kappa_L} |s - s'|.$$

This proves the uniform physical realisation, continuity, and genuine-infeasibility assertions. $\square$

The preceding margin is angular. Its physical size decreases with the radius, so a long exact word need not remain distinguishable at an assigned tolerance. The next lemma chooses a shorter prefix and compares its radial scale directly with that tolerance.

Lemma 12.2 (Forcing a block prefix). *Fix $L \geq 2$ and $1 \leq j \leq L - 1$, and put*

$$M_L = \binom{L-2}{j-1}, \quad c_L = -\frac{j}{L} \log r_0 - \left(1 - \frac{j}{L}\right) \log r_1.$$

For every precision sequence of exponent $\gamma > 0$ and every $0 < t < \min\{1, \gamma/c_L\}$, for all sufficiently large n ,

$$(12.6) \quad r_F(n, \delta_n; Q) \geq M_L^{\lfloor tn/L \rfloor}.$$

Proof. Use all length- L blocks starting with zero, ending with one, and containing j zeros. Every concatenation has runs of length at most L and radial product e^{-Lc_L} per complete block. Put $k_n = \lfloor tn/L \rfloor$ and $m_n = Lk_n$. For each of the $M_L^{k_n}$ initial concatenations, append one fixed allowable block forever. Lemma 12.1 realises these infinite inputs at the same positive radius s_L . Every prefix through time m_n has $P_{v|k} \geq P_{v|m_n} = e^{-c_L m_n}$.

If an arbitrary catalogue input first differs from a witness at $j' \leq m_n$, its preceding states agree with that witness. By (10.6), their radius at $j' - 1$ is at least $C_R^{-1} s_L e^{-c_L m_n}$. The wrong radial factor is at least $r_*/2$, and the alternative angular excess is at least d'_L . Its actual physical excess therefore obeys

$$(12.7) \quad |z_{j'}| - s_{j'} \geq E_L e^{-c_L m_n}, \quad E_L = \frac{s_L r_* d'_L}{2C_R} > 0.$$

The ratio to $2\delta_n$ has logarithmic rate $\gamma - c_L t > 0$. For large n , it violates (10.8), regardless of later reentry. Thus the approximate input must agree with all m_n witness digits. Distinct selected prefixes give distinct actual states: at their first different digit one input is feasible and the other has the proved wrong-control margin. One catalogue input covers at most one of them, proving (12.6). The block length, positive radius, and margin remain fixed throughout this horizon limit. Integer rounding and every $o(n)$ precision fluctuation are absorbed by the strict choice of t . $\square$

13. STOPPING CATALOGUES AND THE MATCHED SPECTRUM

We first construct catalogues by following feasible inputs until the radius is small enough for a common tail. This construction applies under all the standing assumptions. We then prove the matched spectrum using the physical prefix lower bound and the covered-area estimate. The reference tree provides an upper bound; its leaves need not all be realised by actual states.

Lemma 13.1 (A complete radial stopping catalogue). *Put $M = \max\{1, 4\sqrt{2}\Lambda C_R \rho\}$. For $N \geq 0$ and $0 < t \leq 1$, let $\mathcal{L}_P(N, t)$ be the full reference binary tree stopped at the first $P_w \leq t$, or at depth N , whichever occurs first. For $n \geq J$ and $0 < \delta \leq 1$,*

$$(13.1) \quad r_F(n, \delta; Q) \leq 2^J \# \mathcal{L}_P(n - J, \delta/M).$$

The catalogue can retain only transient/leaf pairs whose complete actual feasible initial region is nonempty.

Proof. For every actual initial state choose an exactly feasible infinite input. Its first J letters form a feasible transient ending in Q_ρ ; the subsequent letters reach a reference leaf. Keep every pair realised by an actual state, including singleton regions and boundary states. Before stopping the assigned states are exactly feasible. At a threshold leaf, (10.6) gives

$$s_{|w|} \leq C_R \rho P_w \leq C_R \rho \delta/M \leq \delta/(4\sqrt{2}\Lambda).$$

Since $V = \Lambda s$ on Q , appending the same infinite zero tail and using (9.5) keeps the Euclidean norm at most $\delta/4$ at every later time. The origin belongs to Q , so every constraint holds strictly on that tail. A horizon leaf is exactly feasible through time n , and its later extension is irrelevant. Every constructed input fixes the origin as well. There are at most 2^J transients, proving (13.1). $\square$

13.1. Upper bound for the matched spectrum. For the rest of this section assume $\beta_0 = \beta_1 = \beta$. Then $P_w = \lambda_w^\beta$, $\chi_R = \beta h$, and

$$t_\delta = (\delta/M)^{1/\beta}, \quad b_\delta = -\log t_\delta.$$

Lemma 13.1 uses the tree stopped at $\lambda_w \leq t_\delta$ or at depth $N = n - J$. Put $c_i = -\log a_i$, $c_{\min} = \min_i c_i > 0$, $c_{\max} = \max_i c_i$, and $\Gamma_{n,\delta} = \beta(b_\delta + c_{\max})/n$. A threshold leaf has logarithmic angular cost in $[b_\delta, b_\delta + c_{\max}]$; an unfinished horizon leaf has cost below b_δ . Thus every nonempty leaf of length $\ell \leq n$, containing j zeros with $p = j/\ell$, satisfies

$$(13.2) \quad \ell \chi(p) < b_\delta + c_{\max}, \quad \frac{\ell}{n} \leq \min\{1, \Gamma_{n,\delta}/[\beta \chi(p)]\}.$$

The corresponding term of the binomial law of parameter j/ℓ gives $\binom{\ell}{j} \leq e^{\ell H_b(j/\ell)}$. End-point types have one word. Each leaf type therefore has at most $e^{nS(\Gamma_{n,\delta})}$ words. There are at most $(n+1)^2$ length/type pairs, including a possible empty word. By the actual catalogue construction,

$$(13.3) \quad r_F(n, \delta; Q) \leq 2^J (n+1)^2 e^{nS(\Gamma_{n,\delta})}.$$

No nonempty fixed-radius slice or full terminal image is assumed.

Every function under the maximum defining S is Lipschitz in γ , with common constant at most $\log 2/(\beta c_{\min})$. Their maximum is continuous. Here $\beta b_{\delta_n} = -\log \delta_n + \log M$, so $\Gamma_{n,\delta_n} \rightarrow \gamma$ for every precision sequence. The fixed transient and polynomial factors have zero rate, giving

$$(13.4) \quad \limsup_n n^{-1} \log r_F(n, \delta_n; Q) \leq S(\gamma).$$

This includes $\gamma = 0$ and nonmonotone sequences.

The reference angular leaf weights sum to one, and each is at least $a_* t_\delta$: a threshold leaf has parent weight above t_δ , whereas an unfinished leaf has weight above t_δ itself. This identity concerns a reference tree, not a physical partition. Enlarging a constant for $n < J$ yields the useful coarse bound

$$(13.5) \quad r_F(n, \delta; Q) \leq C_U \delta^{-1/\beta} \quad (0 < \delta \leq 1), \quad r_F(n, \delta; Q) \leq 2^n.$$

13.2. Shorter physical prefixes give the lower bound. Let $\gamma > 0$ and fix $p \in (0, 1)$. Choose $j_L \in \{1, \dots, L-1\}$ with $j_L/L \rightarrow p$, and put

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = L^{-1} \log M_L, \quad \chi_L = \chi(j_L/L).$$

The exact identity and elementary type bounds

$$\binom{L-2}{j-1} = \frac{j(L-j)}{L(L-1)} \binom{L}{j}, \quad \frac{e^{LH_b(j/L)}}{L+1} \leq \binom{L}{j} \leq e^{LH_b(j/L)}$$

give $B_L \rightarrow H_b(p)$ and $\chi_L \rightarrow \chi(p)$. For the lower type bound the j -th term is a largest term of the binomial law with parameter j/L , so is at least $1/(L+1)$. The additional fraction has logarithm $o(L)$ for interior p .

The radial block cost in Lemma 12.2 is now $c_L = \beta\chi_L$. Fix L first and take

$$(13.6) \quad 0 < t < \min\{1, \gamma/(\beta\chi_L)\}.$$

The lemma, which checks the physical margin and every constrained time, gives

$$(13.7) \quad \liminf_n n^{-1} \log r_F(n, \delta_n; Q) \geq tB_L.$$

For fixed L take the supremum over the strict t , next let $L \rightarrow \infty$, and finally take the supremum over $p \in (0, 1)$. Endpoint frequencies have zero entropy, so continuity yields $\liminf n^{-1} \log r_F(n, \delta_n; Q) \geq S(\gamma)$. No witness radius or margin is required to remain uniformly positive as $L \rightarrow \infty$; they are fixed in each horizon limit. For $\gamma = 0$, the bound $r_F \geq 1$ and (13.4) give the zero limit. This proves (9.15).

13.3. Positive-area and fixed near-full initial sets. To complete Theorem 9.3, apply Theorem 9.7 to a fixed positive-area compact K and its positive-area set T . A catalogue covering K has at least $\text{area}(T)/\sup_u \text{area}(T \cap A_u(n, \delta_n; K))$ elements. For fixed $e \in (0, h)$, put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + e} \right\rfloor \right\}.$$

Eventually $m_n \geq 0$, and $\delta_n \leq e^{-(\chi_R + e)m_n}$. Each term on the right of (9.22) is then at most a constant times $e^{-(h-e)m_n}$. Consequently

$$\liminf_n n^{-1} \log r_F(n, \delta_n; K) \geq (h - e) \min\{1, \gamma/(\chi_R + e)\}.$$

Since $\chi_R = \beta h$ here, letting $e \downarrow 0$ gives the lower bound in (9.16); for $\gamma = 0$ the bound $r_F \geq 1$ also suffices. The upper bound follows from $r_F(n, \delta; K) \leq r_F(n, \delta; Q)$ and (9.15). The evaluation of S below will give the claimed common low-side rate.

For $\zeta \in (0, 1)$ take the single near-full set T constructed in Section 11 and put $K_\zeta = T \cup \{0\}$. All exactly feasible length- n words meeting T have, after their J -digit transient, frequency uniformly tending to a_0 . If $d_N \rightarrow 0$ bounds this deviation for tails of length $N = n - J$, there are at most

$$2^J(N+1) \exp \left(N \max_{|p-a_0| \leq d_N} H_b(p) \right) = e^{(h+o(1))n}$$

such words. Every state of T has an exactly feasible prefix, so these words, extended by any infinite tail, cover T through time n . One additional input covers the origin. This gives an upper rate h for every precision sequence, without selecting a strategy. Bound (13.5) also gives upper rate γ/β . The lower bound already proved gives (9.17). The construction of T used all-prefix frequency convergence, removal of null critical pullbacks, and compact inner approximation; it preceded every exponent, precision sequence and horizon. This completes Theorem 9.3 subject to the following elementary evaluation.

13.4. Evaluation of the maximum.

Proof of Corollary 9.4. Reparameterise the maximum by the frequency q of the smaller branch $\alpha = a_*$. This only changes the optimisation variable, not the dynamics or control assignments. Then $\chi = c_b + d_b q$. Write $x = \gamma/\beta$. The entropy inequality

$$c_b + d_b q - H_b(q) = q \log(q/\alpha) + (1 - q) \log((1 - q)/(1 - \alpha)) \geq 0$$

follows from $-\log v \geq 1 - v$ and extends to the endpoints by continuity; equality holds at $q = \alpha$. For $x \leq h$, the objective $H_b(q) \min\{1, x/(c_b + d_b q)\}$ is at most x and equals x at α .

For $h < x < c_b + d_b/2$ put $q_x = (x - c_b)/d_b \in (\alpha, 1/2)$. When $q \leq q_x$ the objective is $H_b(q) \leq H_b(q_x)$. When $q \geq q_x$ it is $xH_b(q)/(c_b + d_b q)$. This ratio decreases for $q > \alpha$: the numerator of its derivative is

$$D(q) = (c_b + d_b q)H'_b(q) - d_b H_b(q), \quad D(\alpha) = 0, \quad D'(q) = (c_b + d_b q)H''_b(q) < 0.$$

The maximum is therefore $H_b(q_x)$. Finally, for $x \geq c_b + d_b/2$, the value at $q = 1/2$ is $\log 2$, the universal entropy maximum. Since $\alpha < 1/2$, $c_b + d_b/2 > c_b + d_b \alpha = h$. The three pieces match at the thresholds, proving (9.18)–(9.19) and the low-side assertion of Theorem 9.3. Subtracting (9.17) gives (9.20). Strict increase of H_b on $(\alpha, 1/2)$ proves the stated strict gap and the exact saturation threshold. $\square$

Corollary 13.2 (Matched linear systems). *If $\beta_0 = \beta_1 = \beta$, the pair (A_0, A_1) has all the conclusions of Theorems 9.3 and 9.7.*

Proof. The angular maps are $(y + a_1)/a_0$ and $(y - a_0)/a_1$. Their feasible domains $[-1, a_0 - a_1]$ and $[a_0 - a_1, 1]$ cover the fibre. Put $t = \max_i t_i < 1$, choose $\Lambda = 1 + 4t/(1 - t)$ and set $v = t(1 + 1/\Lambda) < 1$. Then $V(A_i x) \leq \max\{\max_i r_i, v\}V(x)$ globally. The remaining hypotheses are immediate. $\square$

14. PHYSICAL PROBABILITY AND EXPONENTIAL RELIABILITY

Throughout this section the powers match, $r_i = a_i^\beta$, and (9.11) holds. We first prove probability domination on the whole triangle. The lower bound then uses regions with known positive physical mass, rather than only the individual witnesses needed for the full-state precision spectrum.

14.1. All feasible transient branches. Let J and ρ be as in Section 11, so every exactly feasible J -prefix ends in Q_ρ . Recall D_w , the complete exactly feasible domain, and w^- , the word with its last digit removed.

Lemma 14.1 (Whole-triangle probability domination). *There is a fixed $K_J < \infty$ such that for every $1 \leq |w| \leq J$ and measurable $B_0 \subset \mathbb{R}^2$,*

$$(14.1) \quad \mu(D_{w^-} \cap F_w^{-1} B_0) \leq K_J \text{area}(B_0).$$

Consequently, for every family $\mathcal{V}$ of words of a common length N ,

$$(14.2) \quad \mu\left(\bigcup_{\substack{|\sigma|=J \\ v \in \mathcal{V}}} D_{\sigma v}\right) \leq C_0 \sum_{v \in \mathcal{V}} \lambda_v, \quad C_0 = \max\{1, 2^J K_J C_A \rho^2\}.$$

The constants are independent of N , the word family, tolerance, failure budget, and horizon.

Proof. Compactness gives $d_0 = \min_{i,x \in Q} |\det DF_i(x)| > 0$. On D_{w^-} all inputs to the derivative factors of F_w belong to Q , even when its last output leaves Q . Thus $|\det DF_w| \geq d_0^{|w|}$ there. Cover this compact domain by finitely many local inverse charts for F_w , with compact closures inside regular charts. Change of variables bounds the pulled-back area in each chart by its inverse-determinant supremum times $\text{area}(B_0)$. Sum over the charts and

maximise over the finitely many words of length at most J . All inverse branches meeting the domain are included; their overlap only enlarges the upper bound. No global inverse is used.

An exactly feasible σ sends D_σ into Q_ρ . Feasibility of the tail v places the endpoint in its actual complete region $\mathcal{I}_v$, whose area is at most $C_A \rho^2 \lambda_v$ by (10.7). A possible endpoint at zero has a null pullback by (14.1). Pull back each region and sum over all at most 2^J transients and all v . This proves (14.2). $\square$

This is domination, not equality of word probabilities or a claim that feasible regions form a partition. Applying (14.1) to the first true exit during the transient would bound its area by $JK_J C_Q \delta$. We do not discard this area at failure cost e_n : δ_n can be exponentially larger than e_n . The next construction uses exactly feasible prefixes.

14.2. Two catalogue upper bounds. Fix $\varepsilon > 0$, $n > J$, and $N = n - J$. Keep every exactly feasible length- n prefix σv , with $|\sigma| = J$, whose tail zero frequency p satisfies $I_a(p) \leq \alpha + \varepsilon$. Empty complete regions may be omitted. Append any infinite input to each kept horizon prefix. Every state not covered has an exactly feasible length- n prefix by controlled invariance, and all such prefixes from that state have discarded tails. By Lemma 14.1, its total probability is at most

$$(14.3) \quad C_0 \sum_{I_a(j/N) > \alpha + \varepsilon} \binom{N}{j} e^{-N\chi(j/N)} \leq C_0(N+1)e^{-N(\alpha+\varepsilon)} < e_n$$

for large n . Here $\binom{N}{j} \leq e^{NH_b(j/N)}$ follows from the j -th term of the binomial probability sum. The catalogue has at most

$$(14.4) \quad 2^J(N+1)e^{NL_a(\alpha+\varepsilon)}$$

members. The fixed transient and polynomial factors have zero rate. Continuity of L_a , proved below, gives upper rate $L_a(\alpha)$.

Independently, the full-state stopping construction in Lemma 13.1 gives upper rate $S(\gamma)$. It assigns an actual feasible prefix to each state; after a threshold leaf its common tail satisfies every remaining constraint. We need not combine the two catalogues: choose the cheaper one. Therefore

$$(14.5) \quad \limsup_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \leq \min\{S(\gamma), L_a(\alpha)\}.$$

14.3. A fixed positive mass gives the low-side lower bound. Take one regular compact $T \subset Q$ of area greater than $3/4$ from Theorem 9.7. For fixed $0 < \varepsilon < h$, put

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\beta h + \varepsilon} \right\rfloor \right\}.$$

Then $\delta_n \leq e^{-(\beta h + \varepsilon)m_n}$ and, using $\chi_R = \beta h$, (9.22) bounds the area covered inside T by one input by a fixed multiple of $e^{-(h-\varepsilon)m_n}$. A successful catalogue covers more than $1/2$ area inside T eventually. The union bound and $\varepsilon \downarrow 0$ give

$$(14.6) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \min\{h, \gamma/\beta\}.$$

Since $S(\gamma) = \gamma/\beta$ for $\gamma \leq \beta h$ and $L_a(\alpha) \geq h$, this matches (14.5) on the entire low side. At $\gamma = 0$, the upper bound and $r_{e_n} \geq 1$ give the zero limit. Fixed loss is used here to retain a positive successful mass, not to construct an upper bound with exponentially small failure.

14.4. Positive-area regions forcing complete prefixes.

Lemma 14.2 (Thickened physical words). *For every integer $L \geq 2$ there are $\tau_L, E_L > 0$ such that any family $\mathcal{W}$ of distinct words of a common length m , each admitting an infinite extension with runs at most L , has pairwise disjoint measurable regions $E_w \subset Q$ with*

$$(14.7) \quad \mu(E_w) = \tau_L \lambda_w.$$

Every point of E_w is exactly feasible under w . If $n \geq m$ and $2\delta < E_L P_w$, every input whose actual successful set $A_u(n, \delta; Q)$ meets E_w has prefix w . All constants are fixed before m, n, δ and the word family.

Proof. For each word choose a bounded-run extension $v(w)$. Lemma 12.1 gives a common radius bound $s_L > 0$, continuous centres $y_w(s)$ for $0 < s \leq s_L$, and wrong-control excess $d'_L > 0$. Its construction gives centre angles at distance at least $\Delta_L = 3d_L/4$ from both boundaries at every step, where $d_L = 2a_*^{L+1}$. Put $K_0 = 1/a_* + 3B\rho$ and choose

$$0 < \epsilon_L \leq \min\{1/(2C_A), \Delta_L/(4C_A), d'_L/(4K_0C_A)\}.$$

Define

$$(14.8) \quad E_w = \{(s, sy) : s_L/2 \leq s \leq s_L, |y - y_w(s)| \leq \epsilon_L \lambda_w\}.$$

Fix the initial radius. On the feasible component between the centre and any feasible angle, (10.6) bounds the change in the angle after each prefix $w|j$ by

$$(14.9) \quad |Y_{w|j}(y) - Y_{w|j}(y_w(s))| \leq C_A \epsilon_L \lambda_w / \lambda_{w|j} \leq D_L, \quad D_L = C_A \epsilon_L.$$

If the complete interval $I_w(s)$ ended within the proposed initial interval, continuity would make an initial or intermediate angular constraint attain an endpoint. Its centre is at distance Δ_L , while $D_L \leq \Delta_L/4$, a contradiction. The radii stay positive and below $\rho < 1$, so no radial constraint terminates the interval. Thus the entire interval is feasible. This argument applies even when the complete terminal image is truncated; it does not assume that image is onto.

At each prefix the thickened state and its centre lie on one image graph with logarithmic radial slope at most one. Hence their log radius ratio has absolute value at most $D_L \leq 1/2$, and their radius difference is at most $2\rho D_L$. The alternative angular map changes by at most

$$(1/a_* + B\rho)D_L + B(2\rho D_L) = K_0 D_L \leq d'_L/4.$$

Every thickened point therefore retains a wrong-control excess at least $d'_L/2$ at every prefix, as well as selected-control feasibility. Continuity of the centres makes E_w measurable. Its actual area is

$$2\epsilon_L \lambda_w \int_{s_L/2}^{s_L} s \, ds = \frac{3\epsilon_L s_L^2}{4} \lambda_w; \quad \tau_L = 3\epsilon_L s_L^2/4.$$

Two distinct equal-length words cannot share a point in their regions: at their first difference one selected control would be wrong, contradicting feasibility under both words.

If an arbitrary input first differs at $j \leq m$, the preceding trajectory is exactly the selected one. Its radius is at least $C_R^{-1}(s_L/2)P_{w|j-1} \geq C_R^{-1}(s_L/2)P_w$. The wrong output has radial factor at least $r_*/2$ and angular excess at least $d'_L/2$. Thus

$$(14.10) \quad |z_j| - s_j \geq E_L P_w, \quad E_L = \frac{s_L r_* d'_L}{8C_R} > 0.$$

Either $z - s$ or $-z - s$ has this positive excess. Each function has Euclidean Lipschitz constant $\sqrt{2}$ and is nonpositive on Q , so the distance is at least $E_L P_w / \sqrt{2} > \delta$. The constraint is violated at that time. Later reentry cannot remove the violation. This proves the all-point forcing assertion. $\square$

14.5. Rare physical types and completion of the proof. Suppose $c = \gamma/\beta > h$. Fix an interior frequency p with

$$(14.11) \quad \chi(p) < c, \quad I_a(p) < \alpha.$$

Choose $j_L/L \rightarrow p$ and use the length- L blocks starting with zero, ending with one, and containing j_L zeros. Write

$$M_L = \binom{L-2}{j_L-1}, \quad B_L = L^{-1} \log M_L, \quad \chi_L = \chi(j_L/L), \quad I_L = \chi_L - B_L.$$

The binomial bounds proved in Section 13 give $B_L \rightarrow H_b(p)$, $\chi_L \rightarrow \chi(p)$ and $I_L \rightarrow I_a(p)$. Fix L large enough that $\beta\chi_L < \gamma$ and $I_L < \alpha$. Put $k = \lfloor n/L \rfloor$ and $m = Lk$. The M_L^k concatenations, extended by a fixed allowed block, have runs at most L . Lemma 14.2 gives disjoint regions of equal mass and equal radial product,

$$\mu(E_w) = \tau_L e^{-\chi_L m}, \quad P_w = e^{-\beta\chi_L m}.$$

The physical margin exceeds $2\delta_n$ eventually because its logarithmic ratio has rate $\gamma - \beta\chi_L > 0$. One arbitrary input can then cover points in at most one region. A catalogue of R inputs successful on $1 - e_n$ area must obey

$$(14.12) \quad R \geq M_L^k - \frac{e_n}{\tau_L e^{-\chi_L m}} = M_L^k \left(1 - \frac{e_n}{\tau_L e^{-I_L m}} \right).$$

The parenthesis tends to one, since $\alpha > I_L$. Take the horizon limit with L fixed, then let $L \rightarrow \infty$. This proves

$$(14.13) \quad \liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq \sup_{\chi(p) < c, I_a(p) < \alpha} H_b(p) \quad (c > h).$$

To evaluate this lower bound, let $a = a_* < 1/2$ and use q for the frequency of the smaller branch. Put $c_b = -\log(1-a)$, $d = \log((1-a)/a)$, and $J_a = I_a(1/2) > 0$ in this relabelled notation. Then $\chi(q) = c_b + dq$, $h = \chi(a) = H_b(a)$, and Corollary 9.4 gives the three pieces of S . Strict convexity of relative entropy, together with monotonicity of H_b on $[0, 1/2]$, gives

$$(14.14) \quad L_a(\alpha) = \begin{cases} H_b(q_\alpha), & 0 < \alpha < J_a, \quad I_a(q_\alpha) = \alpha, \quad a < q_\alpha < 1/2, \\ \log 2, & \alpha \geq J_a. \end{cases}$$

This also proves continuity, including at J_a . For $c > h$ set $q_c = \min\{1/2, (c - c_b)/d\}$, and set $q_\alpha = 1/2$ if $\alpha \geq J_a$. Every $a < q < \min\{q_c, q_\alpha\}$ satisfies (14.11). Approaching its endpoint in (14.13) yields

$$\liminf_n n^{-1} \log r_{e_n}(n, \delta_n; Q) \geq H_b(\min\{q_c, q_\alpha\}) = \min\{S(\gamma), L_a(\alpha)\}.$$

Together with (14.5) and (14.6), this proves Theorem 9.1. The common-cost endpoint $c = h$ uses the fixed-mass argument, while the saturation and reliability endpoints use strict types approaching from below. Integer rounding changes m by less than fixed L ; the transient changes N by fixed J . The strict gaps in (14.3) and (14.12) absorb every $o(n)$ fluctuation in both sequences. No constants are required to be uniform as $L \rightarrow \infty$, or over different fixed systems.

Corollary 14.3 (Linear and fixed-failure specialisations). *The matched linear pair has rate (9.12) for every stated pair of sequences. Under the weaker standing null-critical hypotheses, each fixed $e \in (0, 1)$ instead satisfies*

$$(14.15) \quad \lim_n n^{-1} \log r_e(n, \delta_n; Q) = \min\{h, \gamma/\beta\}.$$

Proof. The linear pair is nonsingular on Q , so Theorem 9.1 applies. For fixed failure, choose the single compact set in Theorem 9.3 with area greater than $1 - e$; its catalogue gives the upper rate in (14.15). For the lower rate choose a regular window T of area greater than $(1 + e)/2$. A catalogue with failure at most e covers at least $\text{area}(T) - e > 0$ area there. The fixed-mass proof above, which needs only Theorem 9.7, gives the lower rate. $\square$

15. A TRANSIENT PRESERVING FULL COSTS BUT CHANGING RELIABILITY

We prove Theorem 9.2. The construction alters only the first step from Q . It remains injective and has no positive-area collapse. Its small target regions have physical preimages whose probabilities decay too slowly to omit at the required failure budget.

15.1. A flat increasing Cantor interpolation. Fix the parameters of the theorem and put $\lambda = a_0 a_1$. The inverse reference maps are $f_0(y) = a_0 y - a_1$ and $f_1(y) = a_1 y + a_0$. For the control blocks 01 and 10, write

$$B_0 = f_0 \circ f_1 = \lambda y + a_0^2 - a_1, \quad B_1 = f_1 \circ f_0 = \lambda y + a_0 - a_1^2.$$

Their invariant interval has endpoints and length

$$(15.1) \quad A = \frac{a_0^2 - a_1}{1 - \lambda}, \quad B = \frac{a_0 - a_1^2}{1 - \lambda}, \quad \ell = B - A = \frac{2\lambda}{1 - \lambda}.$$

Indeed $A + 1 = 2a_0^2/(1 - \lambda) > 0$ and $1 - B = 2a_1^2/(1 - \lambda) > 0$, so $[A, B] \subset (-1, 1)$. The maps take this interval to its left and right portions of relative length $\lambda < 1/2$. The corresponding Cantor set $\mathcal{C}$ consists of angles with infinite block controls 01, 10.

On $J_0 = [-1/2, 1/2]$ use the source maps

$$b = 2/5, \quad h_0(y) = by - (1 - b)/2, \quad h_1(y) = by + (1 - b)/2.$$

Their Cantor set $\mathcal{D}$ is null, because the total level- k interval length is $(2b)^k \rightarrow 0$. For $\sigma \in \{0, 1\}^k$ put

$$J_\sigma = h_{\sigma_1} \circ \cdots \circ h_{\sigma_k}(J_0), \quad Z_\sigma = B_{\sigma_1} \circ \cdots \circ B_{\sigma_k}([A, B]).$$

Within each family these closed intervals are separated by positive gaps. Their lengths are b^k and $\ell\lambda^k$, respectively. Both address maps preserve order and are unique, including endpoints. The target Cantor set is null as $(2\lambda)^k \rightarrow 0$.

We now construct a strictly increasing C^2 homeomorphism $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$(15.2) \quad g(y) = y \quad (|y| \geq 1), \quad g([-1, 1]) = [-1, 1], \quad g(J_\sigma) = Z_\sigma, \quad g'(y) = 0 \iff y \in \mathcal{D}.$$

Let $E(t) = e^{-1/t}$ for $t > 0$ and zero otherwise, and set

$$\phi(t) = E(t)E(1 - t), \quad \Sigma(t) = \frac{\int_0^t \phi(v) dv}{\int_0^1 \phi(v) dv} \quad (0 \leq t \leq 1),$$

extended by zero and one outside that interval. The function is smooth, strictly increasing on $(0, 1)$, and flat at both endpoints. Map each point of $\mathcal{D}$ to the point of $\mathcal{C}$ with the same address. On a source gap (u, v) whose corresponding target gap is (U, V) , define

$$(15.3) \quad g(y) = U + (V - U)\Sigma((y - u)/(v - u)).$$

The ordered endpoint correspondence and increasing interpolation give the full interval identity $g(J_\sigma) = Z_\sigma$. At level k , the gap lengths are $(1 - 2b)b^k$ and $(1 - 2\lambda)\ell\lambda^k$. For $j = 1, 2$ the gap derivatives obey

$$(15.4) \quad |g^{(j)}| \leq C_j(\lambda/b^j)^k.$$

Since $\lambda < 2^{-10} < b^2$, both bounds tend to zero with level. Define q_1, q_2 by these derivatives inside gaps and by zero on $\mathcal{D}$. They are continuous, by (15.4) for high-level gaps and endpoint flatness for the remaining finite collection.

These candidate derivatives are actual derivatives. The integral of q_1 over a gap is its target gap length. Nullity of $\mathcal{C}$ and order preservation therefore imply $g(x) = A + \int_{-1/2}^x q_1(y) dy$ on J_0 ; inside a gap include the partial integral of (15.3). Moreover

$$\sum_{\text{gaps}} \int |q_2| \leq C \sum_{k \geq 0} (2\lambda/b)^k < \infty.$$

Each complete-gap integral of q_2 is zero, and a partial-gap integral equals $q_1(x)$. Since $\mathcal{D}$ is null, $q_1(x) = \int_{-1/2}^x q_2(y) dy$. The fundamental theorem of calculus gives $g' = q_1$ and $g'' = q_2$. Thus g is C^2 on J_0 , flat to second order on $\mathcal{D}$, with positive first derivative on all gaps.

For completeness, the exterior joins can be made without altering these properties. On $[L, R]$ with prescribed increment $T > 0$, set $d = \min\{(R - L)/4, T/4\}$ and $v(x) = E(x - L)E(R - x)$. For the left join use $c(x) = 1 - \Sigma((x - L)/d)$, and for the right use $c(x) = \Sigma((x - R + d)/d)$. In either case $\int c \leq d < T$. Integrate the positive interior density

$$(15.5) \quad q(x) = c(x) + \frac{T - \int_L^R c}{\int_L^R v} v(x).$$

For $(-1, -1/2, A + 1)$ start at $g(-1) = -1$; for $(1/2, 1, 1 - B)$ start at $g(1/2) = B$. The identity endpoint has derivatives $1, 0$, and the Cantor endpoint has derivatives $0, 0$. Extending by identity beyond $[-1, 1]$ therefore gives a global C^2 map satisfying (15.2). It is strictly increasing and onto, and $|g(y) - y| \leq 2$ globally. The map is fixed independently of every horizon and both sequences.

15.2. The actual maps and all dynamical assumptions. Set

$$\psi(s) = \frac{E(s - 1/4)}{E(s - 1/4) + E(1/2 - s)}.$$

Define

$$H(s, z) = \begin{cases} (s, z), & s \leq 1/4, \\ (s, z + s\psi(s)[g(z/s) - z/s]), & s > 1/4, \end{cases}$$

and set $F_i = A_i \circ H$. The smooth function ψ is zero on $s \leq 1/4$ and one on $s \geq 1/2$. The quotient is used only away from zero; its needed derivatives are locally bounded at every joining point, and flatness of ψ gives global C^2 maps. Near the origin H is identity, so the required first derivatives agree exactly.

On every fibre the angular map $g_s(y) = (1 - \psi(s))y + \psi(s)g(y)$ is an increasing homeomorphism of $[-1, 1]$ onto itself. Therefore $H(Q) = Q$, and one of the two linear controls is feasible at $H(x)$ for each $x \in Q$. Its radial factor is less than one. This proves controlled invariance including every boundary and the origin.

Put $m_* = \max_i \{a_i^\beta, a_i^{\beta-1}\} < 1$, $\Lambda = 1 + 10m_*/(1 - m_*)$, and $V = \max\{\Lambda|s|, |z|\}$. The bound $|H^z - z| \leq 2|s|$ gives, on the whole plane,

$$V(Hx) \leq (1 + 2/\Lambda)V(x), \quad V(A_i x) \leq m_*(1 + 1/\Lambda)V(x).$$

Consequently the common contraction factor can be taken as

$$\theta = m_*(1 + 1/\Lambda)(1 + 2/\Lambda) \leq m_*[1 + 5/\Lambda] < (1 + m_*)/2 < 1.$$

This applies to arbitrary input tails outside Q as well.

For $s > 1/4$ the determinant is

$$\det DH = 1 - \psi(s) + \psi(s)g'(z/s), \quad \det DF_i = a_i^{2\beta-1} \det DH;$$

for $s \leq 1/4$ the first determinant equals one. Thus the critical set in Q is exactly $\{(s, sy) : 1/2 \leq s \leq 1, y \in \mathcal{D}\}$. It has area zero by Fubini in physical coordinates. Local inverse charts off that null set show that every null target set has a null pullback in Q . No positive-area set is collapsed to a null set. Each fibre map is strictly increasing and identity outside

$[-1, 1]$, so H is globally bijective. It is proper, since its first coordinate is unchanged and $|H^z - z| \leq 2|s|$. Thus it and both F_i are global homeomorphisms. They are not asserted to be diffeomorphisms; the displayed critical points are genuine.

The fixed radial factors satisfy

$$a_1^\beta = (1 - 2^{-10})^{4096} < e^{-4} < 1/4, \quad a_0^\beta < 1/4.$$

After the first input from Q , the radius lies below $1/4$, regardless of whether its angle is feasible. Every subsequent input therefore uses the identity part of H , even along an approximate trajectory that has left Q . For every input and every $j \geq 1$,

$$(15.6) \quad F_u^j x = A_u^j Hx \quad (x \in Q).$$

At time zero both x and Hx belong to Q . It follows that $A_u^F(n, \delta; Q) = H^{-1}A_u^A(n, \delta; Q)$ on Q . Since $H(Q) = Q$, full covers correspond, proving (9.13). Probability covers correspond only after transporting μ to $H_*\mu$, which is absolutely continuous but is not uniform area. This is consistent with measure-transport invariance results; it is not a counterexample to them.

15.3. Positive-area forced families and the strict rate gap. Every infinite sequence of blocks 01, 10 has runs at most two. The reference endpoint-distance identities used in (12.1) give $|y_j| \leq 1 - d$ at all shifted positions, with $d = 2a_*^3$. For a chosen zero, the current angle is at most $a_0 - a_1 - a_0d$, so the alternative one gives at most $-1 - a_0d/a_1$; the symmetric bound holds for a chosen one. Thus every wrong-control angular excess is at least $d_* = (a_*/a^*)d > 0$. This holds on each complete interval Z_σ through its $2k$ -digit block prefix: its endpoints have such bounded-run extensions, and the linear prefix maps are monotone. No symbolic Cantor subset replaces the whole interval in this assertion.

For each $\sigma \in \{0, 1\}^k$ define an actual physical region

$$E_{\sigma,k} = \{(s, sy) : 1/2 \leq s \leq 1, y \in J_\sigma\}.$$

Here $\psi = 1$ and $g(J_\sigma) = Z_\sigma$. The regions are disjoint and each has mass $(3/8)b^k$, so their union has mass $(3/8)(2b)^k$. Every point requires its selected $2k$ -digit control prefix at sufficiently small tolerance. To check this for an arbitrary input, let its first differing digit be $j \leq 2k$. Equation (15.6) and $s \geq 1/2$ give the preceding physical radius at least $\frac{1}{2}\lambda^{\beta k}$: a partial block product before $2k$ is no smaller than the complete product λ^k . The wrong output radial factor is at least a_*^β and its angular excess is at least d_* . Therefore

$$(15.7) \quad |z_j| - s_j \geq c_0 \lambda^{\beta k}, \quad c_0 = \frac{1}{2} a_*^\beta d_* > 0.$$

Its distance from Q is at least this value divided by $\sqrt{2}$. For $k = \lfloor n/2 \rfloor$ and $\gamma = -\beta \log \lambda$, the logarithmic ratio to δ_n has positive rate $-\frac{1}{2}\beta \log \lambda$. Every successful input thus has the full selected prefix, for every point in the region. Later reentry cannot remove the constraint at this first wrong step.

One catalogue input can consequently cover points of at most one region. A catalogue of R inputs with failure at most e_n obeys

$$(15.8) \quad R \geq 2^k \left(1 - \frac{8e_n}{3(2b)^k} \right).$$

Since $-\frac{1}{2} \log(2b) = \frac{1}{2} \log(5/4) < 1/8 = \alpha$ by $\log(1+x) < x$, the parenthesis tends to one. Thus the lower limit is at least $\frac{1}{2} \log 2$. Both strict exponent gaps absorb all the allowed $o(n)$ fluctuations.

Finally $S(\gamma) = \log 2$: at $p = 1/2$ the threshold ratio $\gamma/[\beta\chi(p)]$ is two. For $q = 1/16$,

$$H_b(q) = \frac{1}{4} \log 2 + \frac{15}{16} \log(16/15) < \frac{1}{4} \log 2 + \frac{1}{16} < \frac{1}{2} \log 2,$$

and

$$I_a(q) > \frac{5}{8} \log 2 - H_b(q) > \frac{3}{8} \log 2 - \frac{1}{16} > \frac{1}{8},$$

using $\log 2 > 1/2$. Relative entropy is strictly increasing for $q > a_0$; its sublevel $I_a(q) \leq 1/8$ therefore has right endpoint below $1/16$. Binary entropy is increasing up to $1/2$, so $L_a(1/8) < H_b(1/16) < \frac{1}{2} \log 2$. This proves the strict inequality (9.14) and completes Theorem 9.2.

The example leaves the full-state finite-horizon counts unchanged but changes the probability weights of forced physical regions. Its own accurate reliability spectrum and the existence of its reliability limit are not determined here; the strict lower-limit gap suffices for the obstruction. Absolute continuity and null-set pullback preservation alone do not control exponential tail masses.

16. MISMATCH AND THE COMMON-COST CRITERION

We now return to the standing assumptions with $\beta_0 \neq \beta_1$. The same physical estimates apply, but the angular and radial products no longer have a common-power identity.

Proof of Theorem 9.6. First consider the upper rate on Q . For every $0 < \gamma < \gamma_0$, all branches of the tree in Lemma 13.1 cross $P_w \leq \delta_n/M$ before depth $n - J$ for large n , since

$$P_w \leq e^{-\gamma_0(n-J)} < \delta_n/M \quad \text{at that depth.}$$

The ratio has logarithmic rate $\gamma - \gamma_0 < 0$, so the statement includes every $o(n)$ tolerance fluctuation. For the uncapped radial tree, write $N(t)$ for its leaf count. Replacing a parent by its children preserves $\sum P_w^D = 1$, and each leaf satisfies $r_* t < P_w \leq t$. Hence

$$(16.1) \quad t^{-D} \leq N(t) \leq (r_* t)^{-D}, \quad r_F(n, \delta_n; Q) \leq 2^J N(\delta_n/M).$$

This proves the upper rate γD through an actual full-time catalogue, without asserting that the reference tree is necessary.

For the lower rate put $p_* = r_0^D$, so $1 - p_* = r_1^D$, and $c_R(p) = -p \log r_0 - (1 - p) \log r_1$. Then

$$(16.2) \quad H_b(p_*) = D c_R(p_*).$$

Choose $j_L/L \rightarrow p_*$ and use the blocks in Lemma 12.2, with $c_L = c_R(j_L/L)$ and $B_L = L^{-1} \log \binom{L-2}{j_L-1}$. The binomial calculation in Section 13 gives $B_L \rightarrow H_b(p_*)$ and $c_L \rightarrow c_R(p_*)$. For each fixed L , $c_L \geq \gamma_0 > \gamma$. Every strict $0 < t < \gamma/c_L < 1$ therefore gives lower rate $t B_L$. Take its supremum first, then let $L \rightarrow \infty$. Equation (16.2) gives the lower rate γD . The positive-radius witnesses and wrong-control margins are those of the common physical lemma, not midpoints of assumed full images. This proves the first limit in (9.21).

For any fixed positive-area compact K , take the set T in Theorem 9.7, and set

$$m_n = \min \left\{ n - J, \left\lfloor \frac{-\log \delta_n}{\chi_R + e} \right\rfloor \right\}.$$

For large n it is nonnegative, and $\delta_n \leq e^{-(\chi_R + e)m_n}$. Every term of (9.22) is then bounded by a constant times $e^{-(h-e)m_n}$. Covering the positive area of T implies lower rate $\gamma(h - e)/(\chi_R + e)$, since $\gamma < \gamma_0 \leq \chi_R$. Let $e \downarrow 0$. This proves the asserted universal lower bound $\gamma h/\chi_R$.

For $K = Q$, choose the single set T of area greater than $1 - \zeta$ in the area proof, and put $K_\zeta = T \cup \{0\}$. Keep only stopped pairs in Lemma 13.1 realised on T . They cover it with the complete tails already constructed. Let $q_\delta = -\log(\delta/M)$ and $c_{\max} = -\log r_*$. Each retained tail of length ℓ has

$$q_\delta \leq -\log P_w < q_\delta + c_{\max}, \quad \ell \leq \frac{q_\delta + c_{\max} + \log D_e}{\chi_R - e},$$

by (11.6). At this length, $\lambda_w \geq D_e^{-1} e^{-(h+e)\ell}$, while all reference length- ℓ weights sum to one. There are at most $D_e e^{(h+e)\ell}$ such words. Sum over these lengths and the at most 2^J

transients, and include one input for the origin. For fixed $e \in (0, h)$ this gives

$$(16.3) \quad \log r_F(n, \delta_n; K_\zeta) \leq \frac{h+e}{\chi_R - e} [q_{\delta_n} + c_{\max} + \log D_e] + O_e(\log(1 + q_{\delta_n})) + O_e(1).$$

Because $q_{\delta_n} = \gamma n + o(n)$, letting $e \downarrow 0$ gives upper rate $\gamma h / \chi_R$, matching the lower bound. The set was selected through uniform frequency convergence before e , the precision exponent, the sequence, and the horizon.

Finally let $b_i = r_i^D$. The classical relative-entropy identity is

$$D\chi_R - h = \sum_i a_i \log(a_i/b_i) \geq 0.$$

Concavity of the logarithm proves the inequality, with equality only when $b_i = a_i$ for both labels. Equality would require $\beta_0 D = \beta_1 D = 1$, contrary to mismatch. Thus $D > h/\chi_R$, completing the theorem. $\square$

Proof of Theorem 9.5. If $\beta_0 = \beta_1 = \beta$, Theorem 9.3 gives the common rate $c(\gamma) = \gamma/\beta$ on $(0, \beta h)$. If $\beta_0 \neq \beta_1$, Theorem 9.6 separates Q and one fixed nearly full-area compact set for every $0 < \gamma < \gamma_0$. Any interval adjacent to zero contains such an exponent, contradicting the common-rate property. $\square$

The criterion concerns precisely the first derivatives (9.4), nonsingularity almost everywhere in Q , controlled feasibility, and the common norm inequality. The proof does not extend this necessity statement to arbitrary feedback systems, classify high-precision mismatch rates, or compare a fixed strategy with the optimal catalogue.

17. APPLICATIONS OF THE CRITERION

The following nonsingular example has nonlinear, truncated feasible images. It satisfies the reliability hypotheses and cannot be reduced to the matched linear pair by the stated constraint-preserving simultaneous conjugacies.

17.1. Inward boundary images and empty fixed-radius slices. Let $t = \max a_i^{\beta-1} < 1$, $\Lambda = 1 + 4t/(1-t)$, $v = t(1 + 1/\Lambda) < 1$, and fix

$$0 < c < \min\{1 - a^*, (1-v)/v, 1\}, \quad g(s) = 1 - \frac{cs}{1+s^2}.$$

Consider the single explicit family

$$(17.1) \quad \begin{aligned} F_0(s, z) &= (r_0 s, g(s)t_0(z + a_1 s)), \\ F_1(s, z) &= (r_1 s, g(s)t_1(z - a_0 s)). \end{aligned}$$

Both are global smooth diffeomorphisms: $g \in [1 - c/2, 1 + c/2]$ is positive, the first coordinate has the explicit inverse $s = s'/r_i$, and the second is an affine invertible function of z at that radius. Their first jets are (9.4). For the stated V ,

$$(17.2) \quad V(F_i x) \leq \max\{r^*, (1 + c/2)v\} V(x), \quad \max\{r^*, (1 + c/2)v\} < 1.$$

Thus the norm inequality is checked globally, not inferred from feasible trajectories.

For $0 < s \leq 1$, $a^* < g(s) < 1$. The complete one-step angular domains are

$$(17.3) \quad D_0(s) = [-1, a_0/g(s) - a_1], \quad D_1(s) = [a_0 - a_1/g(s), 1].$$

Their endpoints lie strictly inside $(-1, 1)$, and their overlap has angular width $1/g(s) - 1 > 0$, physical width $s(1/g(s) - 1) = O(cs^2)$ near zero. This proves controlled invariance and the actual overlap geometry. Theorem 9.3 applies.

Yet for every fixed $0 < s_0 \leq 1$, a sufficiently long all-zero word is infeasible for every initial angle. To verify this, write $s_k = r_0^k s_0$ and $q_k = y_k + 1$. Under zero input,

$$(17.4) \quad q_{k+1} = \frac{g(s_k)}{a_0} q_k + 1 - g(s_k).$$

The multiplier is uniformly greater than one, since $g(s) > a^*$, and $q_1 \geq 1 - g(s_0) > 0$ for $y_0 \geq -1$. The sequence eventually exceeds two, violating $y \leq 1$. Monotonicity means that this also holds for the smallest initial angle, hence for all of them. Thus a description in which every word has a nonempty full-image slice does not apply to this example.

This statement concerns $I_{0^m}(s_0)$ at a fixed radius, not the whole physical region $\mathcal{I}_{0^m}$. Choosing the radius smaller can make a given finite word feasible. Indeed any finite word admits a bounded-run infinite extension, and Lemma 12.1 realises that extension at a sufficiently small positive radius. Missing fixed-radius slices must not be confused with removing a word everywhere in Q .

There is also a constraint-preserving conjugacy obstruction. For (17.1), F_0 sends the positive lower edge of Q strictly inside Q , and its upper edge outside; F_1 does the symmetric thing. The image of the top edge has constant radius $r_i < 1$ and crosses the opposite angular boundary exactly once. Thus

$$(17.5) \quad F_i(\partial Q) \cap \partial Q \text{ consists of exactly two points.}$$

In contrast, for any target pair whose zero control sends the whole lower edge of Q into ∂Q and whose one control sends the whole upper edge into ∂Q , both corresponding intersections contain a continuum. A homeomorphism H of the plane with $H(Q) = Q$, simultaneously conjugating the two labelled controls (even allowing a label permutation), would map each intersection in (17.5) onto the corresponding target intersection. Finite versus continuum is impossible. Thus this example is not reducible to such a boundary-edge-preserving pair by a fixed Q -preserving simultaneous control homeomorphism, and in particular not by such a C^1 or bi-Lipschitz map. No assertion about unconstrained coordinate changes or every member of the new class is made.

For example, take $a_0 = 1/3$, $a_1 = 2/3$, $\beta = 2$ and $c = 1/100$. The choice $\Lambda = 9$ in (17.2) gives a global factor at most $67/90 < 1$. Corollary 9.4 evaluates this same inward family at exponent $\gamma = \log 4$:

$$2 \left(\log 3 - \frac{2}{3} \log 2 \right) < \log 4 < \log(9/2), \quad \lim_n n^{-1} \log r_F(n, \delta_n; Q) = H_b \left(\frac{\log(4/3)}{\log 2} \right).$$

This holds for every precision sequence of that exponent. The first inequality follows from $(4/3)^3 > 2$ and the second from $4 < 9/2$. It is an evaluation of the spectrum, not a claim that missing fixed-radius words require a different count.

18. CONSEQUENCES AND SCOPE

Exact viability, fixed-tolerance maintenance, shrinking precision, and exponential reliability measure different aspects of the same initial-state assignment problem. The solid-cone results compute positive exact, outer, and tracking information costs even when all feasible state trajectories converge uniformly. Their common Lyapunov structure rules out state Li–Yorke pairs under arbitrary separate feasible inputs, and separates the state factor and input fibres from entropy retained by the input shift. The normalization at a positive radius explains the catalogue calculation but is singular at the common fixed state.

For all-input norm contraction, fixed tolerance instead has zero outer and tracking rates. Shrinking tolerance retains a nontrivial precision spectrum. Under matched first derivatives and null critical sets, every positive-area compact set shares the low-precision rate, while one fixed nearly full-area set omits the costly rare states. Its selection precedes every finite precision exponent and sequence. The full-triangle spectrum continues to grow above βh and saturates at the distinct threshold $-(\beta/2) \log(a_0 a_1)$. Unequal first-order radial powers separate the costs already near zero within the specified matrix family.

At exponentially decreasing failure, rare states cannot be treated as a fixed area loss. With no critical points on Q , the finite-transient transport bound holds on the whole

probability space. Together with positive-area forced prefixes, it gives the exact precision–reliability rate. In the Cantor transient, ordinary full-state catalogues remain identical to the linear ones, but physical tail probabilities change their exponent. Thus preservation of an ordinary precision spectrum does not imply preservation of reliability. Nonsingularity is a sufficient condition; null critical sets are insufficient for universal probability transfer. No necessary and sufficient probability classification is claimed.

The scope restrictions differ between the two parts. The cone results permit arbitrary transverse dimension and general compact control alphabets, but fix the common radial form; the nonlinear persistence theorem keeps its radial map and first jet fixed. The planar precision results permit angular feedback, folds, vanishing local overlap, and truncated feasible images, but retain binary controls, the prescribed first-order matrices, a triangular controlled invariant constraint, and global contraction toward the origin in a common norm. That norm condition is not contraction of distances between arbitrary points. We make no general claim about dynamics outside the feasible constraint in the cone family.

Uniform physical area is fixed in the reliability result; transporting a measure proves a different statement. A success subset allowed to vary with the horizon is not the single fixed compact set in the initial-set theorems. No causal communication capacity, optimality of a prescribed strategy, high-precision mismatch spectrum, or classification of every compact initial set is deduced. The conjugacy obstruction in the inward example is restricted to Q -preserving simultaneous control maps, and the Cantor example’s exact reliability rate is outside our claims. The appendices give the symbolic-counting and scalar baselines used to distinguish these assertions.

Generative AI disclosure. Generative AI was used extensively in developing mathematical arguments, checking proof chains and related literature, drafting and restructuring the English text, and producing the LaTeX source. These activities do not constitute verification by a human author. Author identity, affiliations, authorship approval and the scope of human mathematical review remain to be supplied and confirmed before any submission.

APPENDIX A. INVARIANT PARTITIONS: WHAT SYMBOLIC COUNTING PROVES

This appendix records the precise implication from an invariant partition and the two realizability failures relevant to the main question. It is supplementary material, not an additional novelty claim.

Definition A.1. An invariant partition is a triple $\mathcal{C} = (\mathcal{P}, \tau, w)$, where $\mathcal{P}$ is a finite Borel partition of Q , $\tau \geq 1$ is an integer, and w assigns to each cell P a control word $w(P) \in U^\tau$ such that every $x \in P$ stays in Q through times $0, \dots, \tau$ under that word. Define the sampled state map

$$G_{\mathcal{C}}(x) = \varphi(\tau, x, w(P)) \quad (x \in P).$$

For $x \in Q$, let $I_{\mathcal{C}}(x) = (P_k)_{k \geq 0}$, where $G_{\mathcal{C}}^k(x) \in P_k$. Write $\Sigma_{\text{real}} = I_{\mathcal{C}}(Q)$ and $\Sigma_{\mathcal{C}} = \overline{\Sigma_{\text{real}}} \subset \mathcal{P}^{\mathbb{N}_0}$.

Proposition A.2. *The map $G_{\mathcal{C}}$ is Borel, Σ_{real} is forward shift invariant, and $\Sigma_{\mathcal{C}}$ is a compact forward shift-invariant set. If N_n is the number of length- n prefixes in Σ_{real} , then*

$$r_{\text{inv}}(n\tau, Q, Q) \leq N_n, \quad h_{\text{inv}}(Q, Q) \leq \tau^{-1} h_{\text{top}}(\sigma|_{\Sigma_{\mathcal{C}}}).$$

If the cells are clopen, $G_{\mathcal{C}}$ and $I_{\mathcal{C}}$ are continuous, $\Sigma_{\text{real}} = \Sigma_{\mathcal{C}}$, and

$$\tau h_{\text{inv}}(Q, Q) \leq h_{\text{top}}(\sigma|_{\Sigma_{\mathcal{C}}}) \leq h_{\text{top}}(G_{\mathcal{C}}).$$

Proof. On each Borel cell the formula for $G_{\mathcal{C}}$ is a continuous finite-time solution map, so the finite pasting is Borel. The identity $I_{\mathcal{C}}(G_{\mathcal{C}}x) = \sigma I_{\mathcal{C}}(x)$ gives forward invariance. Continuity of the shift passes that property to the closure.

For each realized prefix $(P_0, \dots, P_{n-1})$, concatenate $w(P_0), \dots, w(P_{n-1})$ and extend arbitrarily to an infinite sequence. Every initial state having this itinerary prefix stays in Q through time $n\tau$ under that one concatenated word. Thus N_n such controls span Q . Taking closure creates no new finite prefixes: each prefix cylinder is open, so every cylinder meeting the closure already meets Σ_{real} .

The word counts are submultiplicative, $N_{n+m} \leq N_n N_m$, and their normalized logarithms have a limit. The entropy of the one-sided subshift is this limit. Indeed distinct length- n prefixes give a uniformly separated set, while length- $(n + \ell)$ prefixes bound spanning numbers at a scale fixed by ℓ . Finally, for arbitrary horizon k , monotonicity gives

$$r_{\text{inv}}(k, Q, Q) \leq r_{\text{inv}}(\tau \lceil k/\tau \rceil, Q, Q) \leq N_{\lceil k/\tau \rceil}.$$

This proves the sampling factor τ^{-1} without discarding nonsampled horizons.

For clopen cells, finite pasting gives continuity of G_C . Each finite itinerary prefix has clopen preimage, so the itinerary map is continuous in the product topology. Its image is compact and hence closed; it is an onto factor map to Σ_C . The factor entropy inequality proves the last bound. $\square$

If in the last case G_C is also onto and $h_{\text{inv}}(Q, Q) > 0$, the classical theorem [1] applies to G_C . This is a direct consequence of a known theorem under a continuous closed-loop hypothesis, not a new control-specific Li–Yorke principle. On connected Q , a finite partition into nonempty clopen cells has just one cell. Its assigned word can be repeated forever from every state, so $r_{\text{inv}}(n, Q, Q) = 1$ for all n and the exact entropy is zero. Thus that simple clopen route cannot apply to the positive-entropy solid cones.

Remark A.3 (A closure itinerary need not be realized). Let $Q = [0, 1]$, $f(x, v) = 2x + v$, and use the three cells

$$P_0 = [0, 1/2), \quad P_1 = [1/2, 1), \quad P_2 = \{1\},$$

with one-step assigned controls 0, -1 , -2 , respectively. This is an invariant Borel partition. For each n , the state $x_n = 1 - 2^{-(n+1)}$ remains in P_1 for at least its first n sampled positions. Hence the constant sequence $P_1 P_1 \dots$ belongs to the itinerary closure. If it were realized by x , iteration of $x \mapsto 2x - 1$ would give $1 - 2^k(1 - x) \in [1/2, 1)$ for all k , forcing $x = 1$. But $1 \in P_2$, a contradiction. Compactness of Q does not make a Borel itinerary map continuous.

Remark A.4 (One-step graph edges need not concatenate). On the discrete compact space $Q = \{a, b, c\}$, let a single control induce $a \mapsto b$, $b \mapsto b$, $c \mapsto a$. For the partition $A = \{a\}$, $B = \{b, c\}$, the one-step intersection graph has edges $A \rightarrow B$ and $B \rightarrow A$. The graph therefore admits the path ABA . Its only possible initial state is a , whose actual orbit is $a, b, b, \dots$, so that path is unrealizable. Even an irreducible graph of one-step intersections can overcount actual paths. A Markov formula needs a separate path-realization hypothesis; it is not supplied by the existence of each edge. This is why the technical partition assumptions in [23, 22] cannot be omitted.

APPENDIX B. SCALAR BASELINES AND CHANGES OF CONTROL RANGE

B.1. Continuous time: an exact finite-horizon computation. Only this subsection uses continuous time. Consider

$$\dot{x} = \lambda x + u, \quad \lambda > 0, \quad u(t) \in [-\lambda, \lambda], \quad Q = [-1, 1].$$

Controls are measurable essentially bounded functions. The variation-of-constants formula gives a unique solution for all $t \geq 0$. Define $r_{\text{inv}}(T, Q, Q)$ by the finite-catalogue quantifiers of (2.1), with every real $t \in [0, T]$ in place of the integer indices; define exact and outer rates per unit of continuous time.

Proposition B.1. *For every $T \geq 0$,*

$$r_{\text{inv}}(T, Q, Q) = \lceil e^{\lambda T} \rceil, \quad h_{\text{inv}}(Q, Q) = h_{\text{out}}(Q, Q) = \lambda.$$

For any common input and distinct initial states the distance is $e^{\lambda t}|x - y|$, so no unconstrained common-input pair is proximal. The admissible feedback $u = -\lambda x$ induces the identity flow on Q .

Proof. For a fixed input, the time- T solution is an affine function of the initial state with slope $e^{\lambda T}$. Initial states satisfying the final constraint therefore form an interval of length at most $2e^{-\lambda T}$. Covering Q , of length 2, requires at least $\lceil e^{\lambda T} \rceil$ controls.

For $T > 0$, put $\delta = e^{-\lambda T}$ and $N = \lceil 1/\delta \rceil \geq 2$. Choose $c_j = -1 + 2j/(N-1)$, $0 \leq j < N$, and use the N constant inputs $u_j = -\lambda c_j$. The associated solution is

$$\varphi(t, x, u_j) = c_j + e^{\lambda t}(x - c_j).$$

Its final state lies in Q exactly when

$$x \in [(1 - \delta)c_j - \delta, (1 - \delta)c_j + \delta].$$

These intervals cover $[-1, 1]$: the first starts at -1 , the last ends at 1 , and consecutive center gaps are $2(1 - \delta)/(N - 1) \leq 2\delta$. For each covered initial state the trajectory lies between its initial and final values, so remains in Q for the whole real interval $[0, T]$. This proves the upper bound. At $T = 0$, one arbitrary input suffices.

For $Q^\varepsilon = (-1 - \varepsilon, 1 + \varepsilon)$, the final-time lower bound becomes

$$r_{\text{inv}}(T, Q, Q^\varepsilon) \geq e^{\lambda T}/(1 + \varepsilon).$$

The exact catalogue remains an outer catalogue, proving outer rate λ for each fixed $\varepsilon > 0$. Subtracting two solutions with the same input proves their distance formula. Finally $|\lambda x| \leq \lambda$ on Q , and the stated feedback gives $\dot{x} = 0$. $\square$

Two distinct initial states cannot share an infinitely feasible input in this bounded interval, since their distance would become unbounded. The stronger unconstrained calculation in proposition B.1 prevents the common-control conclusion from being a merely vacuous consequence of feasibility. More generally, an autonomous Lipschitz scalar closed-loop vector field preserving a compact interval has convergent trajectories. A nonstationary solution cannot cross an equilibrium by uniqueness, so its derivative has a constant sign along its orbit and it is bounded and monotone. Pair distances therefore have limits, precluding Li-Yorke pairs. This observation is about autonomous scalar feedbacks; it is not a theorem about all time-dependent controllers.

B.2. Interval controls and interior containment in the cone. In the planar digit cone, enlarge U_q to its convex hull $\widehat{U}_q = [-(q - 1), q - 1]$. The exact lower bounds in theorem 6.1 use only the slope of a fixed-word map, and remain valid for this enlarged range. The original digit controls still give the upper bounds. Thus all the exact rates and counts in the planar case remain unchanged. Theorem 3.1(ii) likewise gives the same outer and tracking rates.

For initial states in K_1 , the exact rate is unchanged if strict interior containment is required at every positive integer time. Indeed their ratios $y = z/s$ lie in $[-1/2, 1/2]$. Cover this interval by finitely many open intervals of radius q^{-n} , with centers $c \in [-1/2, 1/2]$, using $O(q^n)$ centers. For each center take $u_0 = -qc$, $u_k = 0$ for $k \geq 1$. These controls lie in $\widehat{U}_q$, since $q/2 \leq q - 1$. For an assigned initial ratio,

$$|y_k| = q^k |y - c| < 1 \quad (1 \leq k \leq n).$$

The radial coordinate remains strictly between 0 and 1, so the ambient states lie in $\text{int } Q$. This proves an upper rate $\log q$, while exact containment gives the matching lower bound.

This assertion is for K_1 ; requiring interior containment from a boundary point of the whole cone is a different question.

Finally, the continuous state map $(s, z) \mapsto (R(s), t(s)z)$ is induced on Q by $\kappa(s, z) = -(q-1)z/s$ for $s > 0$, with $\kappa(p) = 0$. It is feasible and smooth as a state map, although κ is discontinuous at p . This gives a zero-entropy continuous closed loop using the smaller convexified digit range, without the larger range needed for one-step resetting.

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